



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 11322D-001.0
Date: 9/29/2025

Client: County of Hawai'i Carollo Project No.: 203975
Contractor: Nan, Inc.
Submittal Name: Vortex Grit Basin Equipment
Reviewed By: Francisco Martinez / Tanner Howe

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

1. Response to Deviations/Clarifications (Page 20 of 56):
 - a. Acknowledged. A flat separator baffle/annulus ring is acceptable.
 - b. The submittal will not be approved until it is signed and stamped by a Professional Engineer in the State of Hawaii.
 - c. Acknowledged. A flat separator baffle/annulus ring is acceptable.
 - d. Per specification section 11322D paragraph 2.04.B.4.a, a baffle section needs to be removable at a minimum in the locations shown in the drawings. These locations are specified on Drawing 02-M-01-106. The remainder of the baffle may be three pieces. Update accordingly.

1. Bookmarks do not match table of contents. Update bookmarks to match table of contents as required per specification section 01330 – 1.03.C.1..
2. The Contractor shall coordinate all equipment dimensions and elevations with the selected structural layout as indicated in the Drawings. No construction of any portion of the grit basins, grit basin influent channels, grit outlet channels, and grit pump room shall begin until shop drawings for the vortex grit basin equipment are fully approved. In addition, the Grit Inlet Gates will not be fabricated until the vortex grit basin equipment is approved.

2. The grit pump suction pipe elevation is specified in Drawing 02-M-01-503, Key Note 5. The submitted hopper elevation does not conform to the grit suction pipe elevation specified in the Drawings. Confirm the reasoning for setting the hopper elevation as submitted. Adjust the hopper elevation to meet the suction pipe elevation. Provide sufficient clearance between the hopper bottom and grit suction BOP.
3. Confirm whether the fluidizing water connections are in optimal positions as shown in Drawing 02-M-01-503 per section 11322D paragraph 2.303.C. Confirm the fluidizing water pipe CL matches the grit suction CL per Drawing 02-M-01-503.
4. The performance report provided from the Florence Regional WWMF is noted. Confirm whether this installation meets the design criteria listed in the specification for comparable facilities. Note that the Sieve #100 grit reduction does not meet the specified performance. No additional reference installations, with evidence, were provided in the submittal. Per specification section 11322D paragraph 1.05, additional performance verification is required. Ensure the submitted facilities contain the design components and requirements listed in the specification.
5. According to the submittal, the manufacturer's performance guarantee will be supplied with the OEM. Provide the manufacturer's performance guarantee in the equipment resubmittal. The equipment submittal will not be approved until the performance guarantee is provided.
6. According to the submitted shop drawings, all painted items will be painted per the manufacturer's spec. Coat the mixer motor and gearbox per specification 09960. Submit coating plan and data sheets for review.
7. All removable lifting eyes provided for installation shall be retained by the Owner per section 11322D paragraph 2.03.B.
8. No anchor bolt calculations were provided. Per specification section 11322D paragraph 1.04.D.1.b, anchor bolt calculations shall be provided. Provide for the motor/gearbox assembly and annulus mounting.
9. Submittal is missing a marked-up specification section 16222 showing compliance with the 16222 – Low Voltage Motors specification.. Update submittal with a copy of 16222 as required per specification section 01330 – 1.03.G.1.
10. Confirm that the submitted motor explosion proof enclosure has draining as described and required per specification section 16222 – 2.04.A.7.c.
11. Confirm gaskets are provided as listed and required in specification section 16222 – 2.04.A.11.c.
12. Confirm that the motor installed in the hazardous location is in conformance with UL 647 as required per specification section 16222 – 2.04.D.1.a.2)..
13. Provide source testing as required per specification section 16222 – 3.02.B.
14. It is noted that functional testing as required per specification section 16222 – 3.02.C. will be provided under a separate submittal.

☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number: 11322D-001.0
Submittal Title: VORTEX GRIT BASIN EQUIPMENT ☐ For Information Only
TO: County
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec: 11322D	Paragraph: 1.04, 1.05, 2.01, 2.02, 2.03, 2.04, 2.05
Authorized By: Ovivo	Date Submitted: 08/25/2025

Submittal Certification	
Check Either (A) or (B):	
☒ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature:	
Printed Name and Title: Marshall Rucknagel, Project Engineer	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 8/25/2025
SPECIFICATION SECTION # 11322D
SPECIFICATION VORTEX GRIT BASIN EQUI
PARAGRAPH 1.04, 1.05, 2.01, 2.02, 2.03, 2.04
DRAWING na
SUBCONTRACTOR na
SUPPLIER Promark Corp
MANUFACTURER Ovivo

CERTIFIED BY CQCM or Designee : 

SECTION 11322D

VORTEX GRIT BASIN EQUIPMENT

PART 1 GENERAL

1.01 SUMMARY

- A. Section Includes: Induced vortex grit removal systems.
- B. Equipment Tag Numbers:
 - 1. Grit Basin Mixer 1 02-MIX-1210.
 - 2. Grit Basin Mixer 2, 02-MIX-1220.

1.02 REFERENCES

- A. American National Standards Institute (ANSI):
 - 1. B3.15 - ABMA Standard, Load Ratings and Fatigue Life for Ball Bearings.
 - 2. B3.16 - ABMA Standard, Load Ratings and Fatigue Life for Roller Bearings.
 - 3. B16.1 - Standard for Cast Iron Pipe Flanges and Flanged Fittings, 125 pound.
- B. ASTM International (ASTM):
 - 1. A36 - Standard Specification for Structural Steel.
 - 2. A48 - Standard Specification for Gray Iron Castings.
 - 3. A689 - Standard Specification for Carbon and Steel Bars for Springs.
 - 4. A743 - Standard Specification for Castings, Iron-Chromium, Iron-Chromium Nickel, and Nickel-Base Corrosion-Resistant for General Application.

1.03 SYSTEM REQUIREMENTS

- A. Grit Basin Design Requirements:
 - 1. Number of basins: 2.
 - 2. Configuration: 360-degree configuration with sloped inlet tunnel:
 - a. The grit basin shall be sized and constructed in accordance with the manufacturer's approved shop drawings to provide proper retention time and adequate flow pattern to meet required grit capture and removal.
 - b. Flat floor basins are not acceptable.
 - 3. Size: 16 to 16.5 foot diameter.
 - 4. Sloped basin floor.
 - 5. Circular sloped inlet/outlet separator baffle.
 - 6. Unit capacity:
 - a. Hydraulic rated capacity: Minimum 20 million gallons per day (mgd).
 - b. Treatment rated capacity: 13 mgd.
 - 7. Head loss: No more than 3 inches at maximum rated capacity.
 - 8. Water depth in channels at peak flows: As indicated on the Drawings.
 - 9. Mixers are to be constant speed motor driven.
 - 10. Gearbox and Motor shall be rated for Class 1, Division 2, Group D hazardous location.

B. Performance Requirements:

1. Grit basins shall achieve the following removal efficiencies at a flow rate of 13 mgd (per basin). Assume grit specific gravity of 2.65.

Percent Removed	Grit Size, Microns
95	>300
87	≥210 and <300
75	≥150 and <210

2. Grit basins shall achieve the same or better grit removal efficiencies for the full specified treatment range of wastewater flow (1.3-13 mgd).
3. Capacities and Operating Data: As specified above.
4. The grit basins and associated equipment shall be suitable for operation at any flow in the full specified flow range without mechanical problems or grit accumulation on the basin's floor.

1.04 SUBMITTALS



- A. Submit product data and other items specified herein in accordance with Section 01330 - Submittal Procedures.



- B. No construction of any portion of the grit basins, grit basin inlet channels, grit outlet channels, and grit pump room shall begin until shop drawings for the vortex grit basin equipment and dimensions are fully approved.



- C. Product Data: Include the following:
1. Motor characteristics and performance.
 2. Accessories.

D. Shop Drawings: Include the following:

IFC drawing set
will be stamped

1. Shop and erection drawings stamped and signed by a Registered Professional Engineer in the state of Hawaii:

In Stamped IFC drawing set

- a. Include the following structural calculations and details prepared by a Professional Engineer registered in the state where the Project is located:
- b. Seismic and anchor bolt calculations and mounting details for each assembly as required by Section 01850 - Design Criteria and Section 15050 - Common Work Results for Mechanical Equipment.

N/A

N/A

2. Templates or setting plans, with tolerances, for anchor bolts.

3. Erection, installation, and adjusting instructions.

4. Functioning field testing procedure.



5. Dimensional drawings (plans and sections) of the grit inlet channels, grit basin, grit outlet channels, and grit hopper.

Will be provided following
verification of dimensions

6. Manufacturer's certification that they have carefully reviewed all drawings and dimensions of the grit basins, grit basin inlet channels, grit outlet channels, and grit hopper, and that these fully comply with manufacturer's requirements for providing their performance guarantee.

Will be provided 30 days prior
to shipment of equipment

- E. Operation and Maintenance Manuals as specified in Section 01782 - Operation and Maintenance Manuals: Include shop manuals.

In OEM Manual

- F. Manufacturer's performance guarantee.

In OEM Manual G. Manufacturer's installation instructions.

In OEM Manual H. Equipment Manufacturer certification of bearing life.

In OEM Manual
by motor vendor I. Motor shop test results.

In OEM Manual J. Recommendations for short and long-term storage.

- ✓ K. Certified Data and Capability: Include the following:
1. Detailed certified test data, furnished with shop drawings, establishing grit removal performance. Obtain test data from test conducted on operating unit, of same model, or larger and in same service as units to be supplied.
 2. Certified test data to establish that unit meets requirements.
 3. Accepted certified test data does not relieve the Contractor nor manufacturer of equipment from performing additional tests, at no additional cost to the Owner, on installed equipment utilizing wastewater to demonstrate that equipment equals or exceeds minimum requirements as specified.

In separate submittal
from Testing vendor L. Shop and field testing procedures and equipment.

In separate submittal
from Testing vendor M. Removal efficiency testing procedures.

Will be provided
following onsite testing N. Removal efficiency testing data and results.

Will be provided
prior to onsite S/U O. Training lesson plan.

P. Nomenclature labels for equipment.

Will be provided
following successful
S/U and training Q. Warranties.

1.05 QUALITY ASSURANCE

- A. Manufacturer Qualifications:
1. Minimum 10 years of experience in production of equipment substantially similar to the specified equipment.
 2. Minimum of 10 installations with same removal efficiency as listed in Article 1.03 Paragraph B.1.
 3. Four grit removal efficiency tests shall be provided from an installation where similar equipment by the manufacturer is currently in similar service. Each grit removal efficiency test shall meet the removal efficiency in Article 1.03 Paragraph B.1.
 4. Submit evidence of satisfactory operation of basins/equipment similar to the specified basins/equipment in at least four separate facilities in accordance with the following requirements:
 - a. All basins/equipment on the submitted installation list shall utilize the same design for critical components as specified for this project as follows:
 - 1) 360 degree configuration.
 - 2) Sloped inlet tunnel.
 - 3) Sloped basin floor.
 - See exception 3 4) Circular sloped inlet/outlet separator baffle (annulus ring).

- b. The four installations shall satisfy the following requirements:
 - 1) Years in service: Minimum 3 years.
 - 2) Basin inside diameter: Not less than 16 feet.
 - 3) Minimum rated capacity: Not less than 20 mgd.
 - 4) Floor slope: Sloped.
 - 5) Mixer: Same design, configuration, and details as proposed for this project.
 - 6) Service: Municipal wastewater in the U.S. or Canada.
 - c. Multiple grit basins at a plant shall be considered as one installation toward meeting the experience requirements.
- B. Fulfillment of the specified experience requirements shall be a condition of acceptance.
- C. Manufacturer's shop welds, welding procedures, and welders:
 - 1. Qualified and certified in accordance with the requirements of ANSI/AWS D1.1.
 - 2. Visually inspect welding while the operators are making the welds and again after the work is completed. After the welding is completed, hand or power wire brush welds and clean them before the inspector makes the check inspection. Inspect welds under light for surface cracking, porosity, slag inclusions, excessive roughness, unfilled craters, gas pockets, undercuts, overlaps, size, and insufficient throat and concavity. Correct deficiencies until accepted by the Engineer.
- D. One manufacturer shall furnish mixer, drive tube, drive head, and electric motor, as a complete and integrated package for proper coordination and compatibility of equipment.
- E. The mixer, drive tube, and related equipment as indicated on the Drawings are general in nature. Modifications required to accommodate the manufacturer's equipment will be at no cost to the Owner and subject to the Engineer's acceptance.

1.06 DELIVERY, STORAGE, AND HANDLING

- A. Ship equipment, material, and spare parts complete, except where partial disassembly is required by transportation regulations or for protection of components.

1.07 WARRANTY

- A. As specified in Section 01783 - Warranties and Bonds.

1.08 SPARE PARTS: PROVIDE THE FOLLOWING SPARE PARTS

- A. Grit basin spare parts:
 - 1. One motor.
 - 2. Two change in lubrication for each gear box supplied.

- B. Spare parts shall be packaged with labels indicating the contents of each package and shall be delivered to the Owner as directed.
- C. Include a price list and name, address, and telephone number of local supplier.
- D. Special Tools:
 - 1. Any special tools required to assemble, disassemble, repair, and/or maintain any item of mechanical equipment shall be furnished with the equipment:
 - a. Special tools shall include any type of tool that has been specifically made for use on an item of equipment for assembly, disassembly, repair, and maintenance.
 - b. When special tools are provided, they shall be marked or tagged, and a list of such tools shall be included with the maintenance and operation instructions describing the use of each marked tool.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. One of the following or approved equal:
 - ✓ 1. Ovivo: Jones + Attwood Products, Jeta Grit Trap 360.
 - 2. Veolia: John Meunier, Mectan V.

2.02 EQUIPMENT

See exception 1

- ✓ A. Provide induced vortex grit removal system with mixer and circular sloped separator baffle for use, as indicated on the Drawings and specified herein capable of removing grit from raw wastewater in a storage hopper where it can be pumped out for further dewatering by separate equipment.

2.03 GENERAL

- ✓ A. Comply with applicable requirements of Section 15050 – Common Work Results for Mechanical Equipment:
 - 1. Section 15050 – Common Work Results for Mechanical Equipment Article 2.01B Rotodynamic Analysis and Testing Vibration is not required for this equipment.

2.04 COMPONENTS

- ✓ A. Multi-Blade Mixer and Drive Tube:
 - 1. Mixer and Blades:
 - a. Minimum 1/4-inch thick, Type 316L stainless steel.
 - b. Adjustment to the performance of the mixer shall be available by resetting the vertical position on the impeller drive tube.
 - 2. Drive Tube:
 - a. Minimum 1/4-inch thick, Type 316L stainless steel.
 - b. Maximum dynamic deflection of 1/4 inch per 10 feet of length.
 - c. Rotating speed, maximum 13 rpm.

- d. Flange-mounted to the bull gear and extends down through the mixer to allow its attachment.
- e. No other attachments to the drive tube shall be permitted, to prevent interference with the balance and free rotation of the mixer.

- ✓ B. Inlet/Outlet Separator Baffle (Annulus Ring):
- 1. Each unit shall include a circular sloped inlet/outlet separator baffle dividing the unit top chamber in 2 distinct zones.
 - 2. The separator baffle shall provide a physical barrier for the ascending flow patterns of the vortex unit. This configuration shall enhance the unit grit capture, while providing effective 360 degrees arrangement.
 - 3. The baffle shall be anchored to the periphery of the grit removal basin. Anchors shall be supplied and installed by the Contractor.
 - 4. Baffle construction:
 - a. Removable at a minimum the locations shown on the drawings. Removable section(s) shall include lifting eyes.
 - See exception 4 b. Constructed of minimum 4 pieces.
 - c. Type 316L stainless steel.
 - d. Minimum 1/8-inch thick plate, with minimum 3/16-inch support angles.
 - e. Maximum allowable deflection: L/180.
 - f. Designed suitable for a person to stand on it and a minimum live load of 20 psf.

- ✓ C. Fluidizing water:
- 1. A fluidizing system that injects water into the bottom of the grit hopper to keep collected grit in suspension, using non-potable plant utility water, shall be provided by Contractor.
 - a. Manufacturer shall identify the ideal connection location(s) for fluidizing water at the bottom of the grit hopper. Submit for approval.

- ✓ D. Utility water requirements per fluidizing system:
- 1. Flow rate: minimum 35 gpm.
 - 2. Pressure: minimum 55 psig.

- ✓ E. Motors:
- 1. Integral part of the geared motor that is directly shaft mounted to the sprocket within the drive head.
 - 2. As specified below and in Section 16222 - Low Voltage Motors up to 500 Horsepower:

Horsepower, minimum	1
Synchronous speed	1,800 revolutions per minute
Voltage	460 volts
Phases	3 phase
Frequency	60 Hertz
Service factor	1.15
Enclosure	TEFC, Suitable for Class 1 Division 2 hazardous environment
Temperature rise	Class B
Insulation	Class F

3. The indicated motor horsepower is estimated. Submit calculations and data to substantiate proposed motor horsepower. If larger horsepower is required, all costs related to electrical system modifications and installations shall be borne by the Contractor.
4. Accessories – as specified in Section 16222 – Low Voltage Motors up to 500 Horsepower, and as indicated on the Drawings.



F. Drive Head:

1. Drive Enclosure: Heavy duty cast iron ASTM A48 Class 30 case or Type 316 stainless steel:
 - a. Base and cover securely bolted together to form a composite unit.
 - b. The base section shall support the ball bearing slewing ring or turntable bearing and shall be designed to prevent distortion of the bearing.
2. Drive System:
 - a. Rotation shall be achieved by means of a bull gear wheel securely bolted to the rotating member of the slewing ring and driven by a steel drive pinion. The bull gear shall be heavy-duty Meehanite flake graphite iron or equivalent.
 - b. An alloy cover shall support the geared motor and a location in the cover enable the correct running centers of the gear wheel and pinion to be achieved.
 - c. The bull gear shall have an opening for the drive torque tube to the mixer.
3. Bearings: Minimum L10 life rating of 100,000 hours.
4. Gears: Heat-treated steel and a service factor of 5.0.
5. Pinion: Service factor of 5.0.
6. AGMA Rating: Class IV for 24 hour a day operation.



G. Hardware and Fasteners:

1. Type 316 stainless steel.

2.05 FINISHES



A. Clean stainless steel components in the either of the following methods:

1. Pickle and passivate all stainless steel components in the factory as follows:
 - a. Pickling and passivation per ASTM A380.
 - b. Wash welds with acid solutions.
 - c. Inspect visually for degradation.
2. Abrasive blast all stainless steel components in the factory as follows:
 - a. Blasting shall be done to remove all visual trace of ferrous oxide or scaling. Acceptability of method will be determined by Engineer based on 1 foot by 1 foot coupon test submitted by manufacturer.
 - b. Blasting material: Glass beads, iron-free silica, or alumina sand free of iron or iron oxide per ASTM A380.
 - c. Clean all components after blasting to remove grit resulting from blasting process.



B. Do not paint stainless steel and plastic components.



C. Surface preparation, factory prime and field finish coat of steel components as specified in Section 09960 - High-Performance Coatings.

PART 3 EXECUTION

3.01 INSTALLATION

- A. No construction of any portion of the grit basins, grit basin influent channels, grit outlet channels, and grit pump room shall begin until shop drawings for the vortex grit basin equipment and dimensions are fully approved.
- B. Install items in accordance with manufacturer's printed instructions, as indicated and specified.
- C. Install equipment on concrete pads and align thereon.
- D. Coat steel components as required by Section 09960 - High-Performance Coatings.
- E. Adjust equipment to operate at optimum performance.
- F. At final completion of project, Contractor shall replace lubrication for each gear box.

3.02 FIELD QUALITY CONTROL

- A. Testing, Training, and Startup: As specified in Section 01756 - Commissioning and as described below.
- B. Coordinate field service work with the manufacturer's representative, Owner, and Engineer prior to initiating such work.
- C. Testing and startup:
 - 1. Performance Testing to be performed during Commissioning Phase as part of Installation Testing specified in Section 01756 - Commissioning:
 - a. Conduct performance tests to substantiate conformance with the specified performance requirements. Submit detailed testing procedures for the Engineer's review and approval.
 - b. The manufacturer shall bear all costs associated with performance testing.
 - c. Conduct tests in presence of the Engineer and Owner. The tests shall be conducted by an independent testing laboratory or firm approved by the Engineer, their services paid for by the manufacturer:
 - 1) Conduct a performance test with normal plant influent wastewater flow during the peak diurnal flow period. Testing is allowed during either dry weather or wet weather. Test one grit basin to demonstrate grit capture performance. All plant influent flow should be routed through this grit basin for the test.
 - 2) Provide and add seed sand (Granusil Mineral Fillers, Gradation 5020) for the grit capture test in accordance with the equipment manufacturer's instructions. Add sufficient quantity of the seed sand into the grit basin influent channel to achieve the required sample size. Samples shall be taken at grit basin influent channel and grit basin effluent channel.

- 3) During the grit performance test, all flow must be diverted through the unit being tested, provided it remains within the peak treatment flow limits of the unit. Flow must be monitored and maintained at the highest flow possible under dry weather conditions.
- 4) Grit must be collected over a period of time to allow for an accumulation of sufficient grit to obtain workable sample weights. Each sample from the influent channel shall be not less than 150 grams, or the size required by the manufacturer for proper testing, whichever is higher. For the purposes of this test, grit shall be defined as any inorganic mineral matter with a specific gravity of 2.65.
- 5) A minimum of six samples shall be taken (three from the grit basin influent channel and three from the grit basin effluent channel). One sample from the influent channel and one sample from the effluent channel shall be taken each day for three consecutive days:
 - a) Collect samples using either an isokinetic vertically-integrated sampler or using 1-inch grit sample probes provided by the Grit Equipment manufacturer:
 - (1) 1-inch sample probes shall be monitored during each test run to ensure that there is no pluggage and that the full flow is obtained.
 - b) Sample probes will be located in the influent and effluent channels and routed to large containers to be utilized as settling basins.
- 6) The influent and effluent samples shall each be fed tangentially and continuously into separate settling barrels with an overflow rate of no more than 3.2 gpm per square foot of surface area, which is intended to capture of 50 micron and larger, 2.65 SG grit particles in the samples. At the end of the test interval, allow a quiescent settling period of 2.3 minutes per foot of barrel depth and then decant the barrels.
- 7) At the termination of each test run, the supernatant shall be siphoned from the top of the settling basin until approximately 1 inch of liquid remains in the bottom. All grit shall be carefully washed from the settling basin to ensure 100 percent capture.
- 8) Carefully transfer the washed grit into a container, pouring off any excess wash water and using caution to retain all grit. The container must be sealed for shipment to an off-site laboratory for burning and analysis.
- 9) The settled solids samples shall be analyzed to determine Total Solids (TS), Volatile Solids (VS) and Fixed Solids (FS). Laboratory testing of the dewatered grit samples shall include wet sieve analysis of the settled solids using 75, 106, 150, 212, and 300 micron (200, 140, 100, 70 and 50 mesh) sieves. Measurement of the solids retained on each sieve including pan dry weights before and after muffle furnace burning of the organics shall be documented. The analytical testing shall be performed in accordance with established protocols as follows:
 - a) Total Solid per Standard Methods 2540G.
 - b) Volatile Solids per Standard Methods 2540G.
 - c) Sand Equivalent Size (SES).

- 10) The percent grit removal of grit particles 75, 106, 150, 212, and 300 micron shall be determined from SES and sieve analysis data collected by the test procedures on each of the three samples taken concurrently at the sampling ports.
 - 11) Submit a final report summarizing all test data and results. The report shall include a mass balance calculation of the grit removal efficiency of the grit basin. It shall allow for ± 5 -percent variation in the removal percentages to allow for variation in site conditions and sampling errors.
 - d. In the event that the equipment does not meet the performance testing requirements, the manufacturer shall make all necessary changes and retest at no additional cost to the Owner.
 - e. If the second test does not successfully demonstrate compliance with the specified performance requirements, the manufacturer shall take additional, concrete steps to rectify the situation until the specified performance is achieved.
 2. Upon completion of installation and successful performance testing, the Manufacturer shall provide certificate of installation and functional compliance.
- D. Require manufacturer's representative to perform the following services as described below and as specified in Section 01756 - Commissioning. The specified durations are the minimum required time on the job site. Additional services and/or longer durations shall be provided as needed at no cost to the Owner to meet the required quality of work. Work to be done in a minimum of 4 trips:
1. Installation assistance, 1 trip; 2 workdays:
 - a. Supervise location of anchor bolts, setting, leveling, alignment, field erection and installation of equipment.
 - b. Coordinate electrical and miscellaneous utility connections.
 - c. Provide additional assistance as required.
 2. Installation inspection: 1 trip; 2 workdays.
 3. Start-up/performance testing (Functional Testing) assistance, 1 trip; 5 workdays:
 - a. Prior to start-up, the equipment shall be inspected for proper alignment, operation, and satisfactory performance.
 - b. Provide additional start-up/testing assistance as required.
 - c. Provide Certificate of Installation and Functionality Compliance per Section 01756 - Commissioning.
 4. Training: Provide training as follows:
 - a. Operations Training: 2 hours class, 2 sessions.
 - b. Mechanical Maintenance Training: 4 hour class, 1 session.
 - c. Electrical Maintenance Training: 2 hours class, 1 session.
 5. Final Acceptance Checkout: 1 workday (trip may be combined with training).

END OF SECTION

Ovivo USA, LLC

4246 S. Riverboat Rd., Suite 300
Salt Lake City, Utah
84123
USA

Telephone: 801.931.3000
Facsimile: 801.931.3080

www.ovivowater.com



August 22, 2025

Submittal Rev.: B

SUBMITTAL INFORMATION PREPARED FOR:

Nan, Inc.

636 Laumaka Street
Honolulu, HI 96819

Project Name:
Hilo WWTP, HI

Project Location:
Honolulu, HI

Equipment Type:
Vortex Grit Removal System

Specification Section:
11353A

Consulting Engineer:
Carollo

Customer Order No.: 24077-00036

OVIVO Sales Order No.: CSW0001868-02

Ovivo USA, LLC

4246 S. Riverboat Rd., Suite 300
Salt Lake City, Utah
84123
USA

Telephone: 801.931.3000

Facsimile: 801.931.3080

www.ovivowater.com



Your Purchase Order No: 24077-00036

August 22, 2025

**Nan, Inc.
636 Laumaka Street
Honolulu, HI 96819**

ATTN: Jyun-Cheng Jhuo

Enclosed are documents pertaining to your order. A description of the documents and their submittal codes are shown on the attached "Submittal Information and Conditions" sheet.

Please note the "Required Return Date" highlighted.

After you review the documents submitted "For Approval" or "Revised for Approval," please make your approval comments, sign one copy of each document, and return them as specified below.

Electronic Copies: Monica.Bonner@ovivowater.com

**Hard Copies: Ovivo USA, LLC
4246 S. Riverboat Rd., Suite 300
Salt Lake City, UT 84123
ATTN: Monica Bonner**

Documents being submitted as "Revised for Final," "Final Certified," or "For Information Only" need not be returned and are submitted for your records and distribution.

Sincerely,

Ovivo USA, LLC

Monica Bonner

Project Manager

CC: Promark Corporation

Ovivo USA, LLC

4246 S. Riverboat Rd., Suite 300
Salt Lake City, Utah
84123
USA

Telephone: 801.931.3000
Facsimile: 801.931.3080

www.ovivowater.com



SUBMITTAL INFORMATION AND CONDITIONS

Submittal Date:	22-Aug-2025	Rev.:	B	Indicates Revision
------------------------	--------------------	--------------	----------	---------------------------

Copies

Submittal In Response To: Customer Order
Required Return Date: 19-Sep-2025
Customer P.O. Number: 24077-00036
OVIVO Sales Order Number: CSW0001868-02

Print: -
Email: N/A
CD's: N/A
Other: N/A

**SUBMITTAL
CODE
KEY**

- | | |
|-----|-----------------------------|
| XXX | 1. For Approval |
| | 2. Revised for Approval |
| | 3. Revised for Final Record |
| | 4. Final Certified |
| XXX | 5. For Information Only |

ENTER

DOCUMENT NO.	DESCRIPTION	CURRENT REV. LEVEL	SUBMIT CODE
(2) Pages	Cover Letter	B	5
(1) Page	Specification Deviations & Exceptions	A	5
(28) Pages	JETA Cutsheets	A	5
(1) Page	Certified Data and Capability	A	5
CSW1868-201	General Arrangement	A	1
Indicates revision from previous submittal		Engineering Change Order #:	ECO-S-025133
REMARKS AND EXPLANATION			

Ovivo USA, LLC
4246 S. Riverboat Rd., Suite 300
Salt Lake City, Utah
84123
USA

Telephone: 801.931.3000
Facsimile: 801.931.3080

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TABLE OF CONTENTS

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COVER LETTER..... 1

SPECIFICATION DEVIATIONS & EXCEPTIONS.....2

JETA CUTSHEETS.....3

CERTIFIED DATA AND CAPABILITY.....4

DRAWINGS.....5

CSW1868-201: GENERAL ARRANGEMENT

SUBMITTAL DATA SHEET

PROJECT NAME _____ Hilo WWTP, HI

LOCATION _____ Honolulu, HI

CUSTOMER _____ Nan, Inc.

CUSTOMER P.O. NUMBER _____ 24077-00036

OVIVO S.O. NUMBER _____ CSW0001868-02

EQUIPMENT DESCRIPTION _____ Vortex Grit Removal System

SPECIFICATION SECTION _____ 11353A

CONSULTING ENGINEER _____ Carollo

ITEM DESCRIPTION UNDER THIS COVER:

COVER LETTER..... (2) Pages

CSW0001868-02 ~ For Information Only ~ 8.22.2025

Ovivo USA, LLC
4246 S. Riverboat Rd., Suite 300
Salt Lake City, Utah
84123
USA

Telephone: 801.931.3000
Facsimile: 801.931.3080

www.ovivowater.com



August 22, 2025

Nan, Inc.
636 Laumaka Street
Honolulu, HI 96819

ATTN: Jyun-Cheng Jhuo

Project Name: Hilo WWTP, HI
Customer Order No.: 24077-00036
Ovivo Sales Order No.: CSW0001868-02
Equipment Description: Vortex Grit Removal System

Dear Jyun-Cheng,

I am the Ovivo USA project manager for the portion of the above referenced project. I will be your first point of contact at Ovivo for submittal and technical questions, scheduling, and all project matters regarding this equipment. You may reach me directly at (385) 564-3272 or via email at Monica.Bonner@ovivowater.com.

We thank you for your order and look forward to a mutually beneficial project. Enclosed with this cover letter is the initial submittal for review and subsequent approval of the equipment. The following schedule is anticipated for the project:

- | | |
|---|------------------|
| ● G.A. Drawings: Submitted for Approval | 8/22/2025 |
| ● Customer Return of Approvals | 9/19/2025 |
| ● O & M Manuals: Sent to Customer | 2/4/2026 * |
| ● Anticipated Equipment Ship Date | 3/6/2026 * |

* These dates are based upon receiving stamped Approved / Approved as Noted drawings by 9/19/2025. Any delay in receiving approved drawings by this date will require a change in the Equipment Ship Date, per Ovivo's Proposal to ship 24 weeks after receipt of approvals, and each affected date will be rescheduled as necessary at the time approved drawings are received.

Improvement upon this anticipated ship date is possible and I will keep you informed of such, but if you wish to make changes to this timetable, please notify me in writing within 10 days of receiving this correspondence and Ovivo will try making adjustments as resources will allow.

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84123
USA

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Subsequent to this cover letter is Ovivo's scope of supply for the above referenced equipment, followed by specification deviations and/or clarifications regarding the submittal package. Dimensions and elevations shown on the submittal drawings have been taken from the contract drawings prepared by the Consulting Engineer. All elevations and dimensions are to be field verified by the Customer. Dimensions and elevations that are clouded or marked with an asterisk (*) are in question, are of particular concern, or deviate from the contract drawings and should be particularly noted. Any obstructions that could potentially interfere with the intended operation of the equipment or dimensional discrepancies found are to be reported to Ovivo in a timely manner so that they may be incorporated into the final design of the custom equipment being provided. Unless indicated otherwise on the return approval drawings, the dimensions and elevations as currently shown will be considered by Ovivo to be correct. Any changes to drawings subsequent to receipt of the approved submittal package will cause Ovivo to incur substantial costs for engineering and material changes. A change order compensating Ovivo for these increased costs will be required. For this reason, I strongly recommend that care be taken to verify all dimensions and elevations during the submittal process.

The submittal package provided herein is complete and accurate to the best of my knowledge. All products submitted meet requirements of the standards referenced. I hope to resolve any concerns or questions in a timely manner so as to expedite the approval process and move the equipment into detailing and fabricating. If Ovivo can do anything to help accelerate the approval process, please let me know.

If you have any questions or concerns regarding this correspondence please contact me at (385) 564-3272, or via email, Monica.Bonner@ovivowater.com, at your convenience.

Sincerely,
Ovivo USA, LLC

A handwritten signature in black ink that reads "Monica R. Bonner". The signature is written in a cursive, flowing style.

Monica Bonner
Project Manager

SUBMITTAL DATA SHEET

PROJECT NAME _____ Hilo WWTP, HI

LOCATION _____ Honolulu, HI

CUSTOMER _____ Nan, Inc.

CUSTOMER P.O. NUMBER _____ 24077-00036

OVIVO S.O. NUMBER _____ CSW0001868-02

EQUIPMENT DESCRIPTION _____ Vortex Grit Removal System

SPECIFICATION SECTION _____ 11353A

CONSULTING ENGINEER _____ Carollo

ITEM DESCRIPTION UNDER THIS COVER:

SPECIFICATION DEVIATIONS & EXCEPTIONS..... (1) Page

CSW0001868-02 ~ For Information Only ~ 8.22.2025

CSW1868 Hilo WWTP - Specification Deviations & Clarifications

	Specification/Drawings	Section:	Requirement:	Exception/ Questions (Ovivo):	Reviewer Response	Ovivo response
Number	Number:					
Exceptions and Deviations - Mechanical						
1	SECTION 11322D	1.03 - A.5.	Circular sloped inlet/outlet separator baffle.	Exception: Ovivo will supply our standard circular flat separator baffle/annulus ring. We gurantee performance with past project experiences.		
2	SECTION 11322D	1.04 - D.1.	Shop and erection drawings stamped and signed by a Registered Professional Engineer in the state of Hawaii:	Clarification: Ovivo will supply the General Arrangement and General Erection drawings but will not provide detailed fabrication drawings for each component. However all the drawings will still be stamped and signed by a registered PE in Hawaii.		
3	SECTION 11322D	1.05 - A.4.a.4)	Circular sloped inlet/outlet separator baffle (annulus ring).	Exception: Same to exception 1. Ovivo will supply our standard circular flat separtor baffle/annulus ring.		
4	SECTION 11322D	2.04 - B.4.b.	Baffle construction: Constructed of minimum 4 pieces.	Exception: Ovivo's standard annulus ring is constructed by 3 pieces of plate, 3 pieces of rim and 3 pieces of suport angle.		

SUBMITTAL DATA SHEET

PROJECT NAME _____ Hilo WWTP, HI

LOCATION _____ Honolulu, HI

CUSTOMER _____ Nan, Inc.

CUSTOMER P.O. NUMBER _____ 24077-00036

OVIVO S.O. NUMBER _____ CSW0001868-02

EQUIPMENT DESCRIPTION _____ Vortex Grit Removal System

SPECIFICATION SECTION _____ 11353A

CONSULTING ENGINEER _____ Carollo

ITEM DESCRIPTION UNDER THIS COVER:

JETA CUTSHEETS..... (28) Pages



HIGH PERFORMANCE, COMPACT GRIT REMOVAL UNIT

Excellent grit removal performance

Compact design for reduced footprint

Fully automatic, low headloss system

Integral grit washing action

Suitable for above ground or below
ground installation

No submerged maintenance points

Rejects lighter grits and organics

Available in 270 and 360 Designs



Interested in
prolonging the life
of your downstream
equipment?

Call 1-855-GO-OVIVO to
speak with an Ovivo
Expert.

JONES + ATTWOOD®

JETA® GRIT TRAP

THIS WELL ESTABLISHED METHOD OF GRIT REMOVAL HAS AN INSTALLATION LIST OF OVER 2,500 UNITS WORLDWIDE.

Grit removal is a prerequisite of wastewater treatment to make the ongoing effluent a manageable and consistent medium when it enters the next phase of treatment.

Screening or comminution of the main flow would ideally be performed prior to grit removal, but in cases where grit is to be removed from entirely raw sewage, additional washing and classification is necessary before dewatering. The Jeta® grit trap provides the highest possible grit removal performance in both these circumstances.



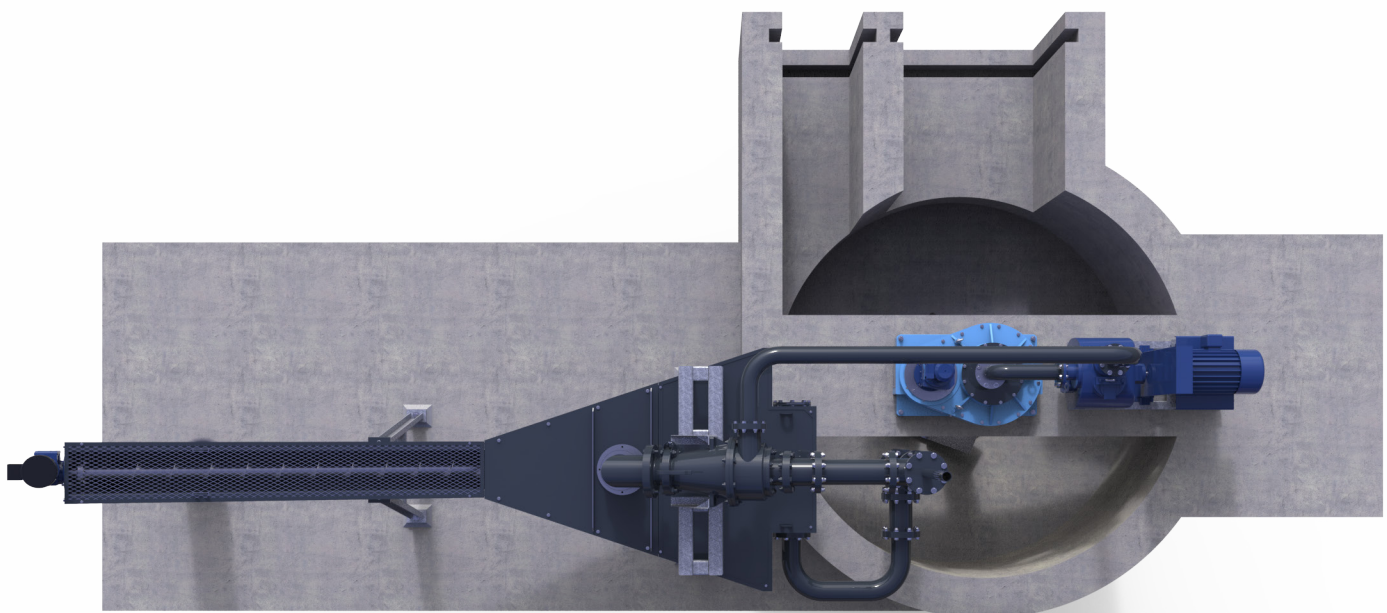
THE JETA GRIT TRAP IS THE RESULT OF THOROUGH DESIGN AND DEVELOPMENT BY J+A.

Ovivo is now proud to deliver all J+A products worldwide.

- Low cost
- Low energy usage
- Simple, slow rotating mechanism
- Easy to maintain
- Capable of 95% capture of 100 mesh and larger Grit

GRIT REMOVAL THROUGH TOP OR BOTTOM MOUNT RECESSED IMPELLER GRIT/SOLIDS PUMP

The J&A Jeta is designed to work in conjunction with Ovivo grit classifiers.



HOW IT WORKS

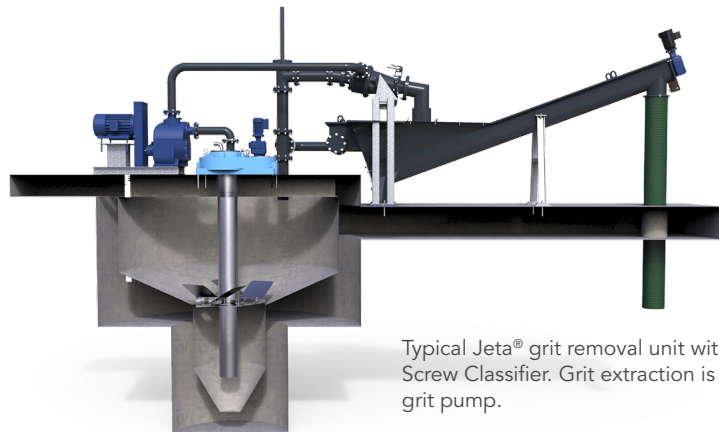
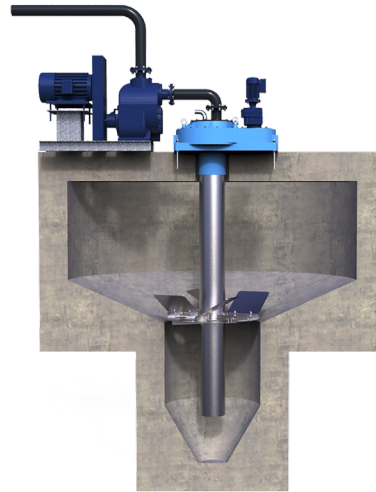
The Jeta grit trap is a compact, circular trap located in the main stream of the inlet system. Effluent enters tangentially, flows around the tank and exits parallel to the inlet. Grit settles within the lower hopper and is transferred to the washing, classification and dewatering plant by a pumping system. With its sloping sides and specially arranged impeller, the Jeta grit trap provides maximum separation of heavy mineral solids (grit) while rejecting larger and lighter solids, which remain in the water flow. Removing these heavy solids protects the rest of the downstream plant from wear, ensuring greater efficiency and reduced maintenance for the rest of the plant.

Two positive zones are set up within the inlet chamber. The outer zone is quiescent, and thus allows the grit and other light solids to settle on the inclined floor of the chamber.

The impeller's predetermined speed produces radial forces that allow the grit to fall into the collection hopper. This force also acts on the lighter solids, preventing them from entering the hopper. They are thus retained in the water flow. The inner zone is made to spiral, and provides an uplift current for lighter solids. These elevated solids then exit the trap. The classifying effect produced can be controlled by adjusting the impeller speed.

The most popular installed arrangement for the Jeta grit trap is shown in the photograph (right).

The circular tank is constructed in concrete and the bridge is cast on site. The impeller drive mechanism is located on the bridge, and can be either belt or gear driven. The centrally mounted airlift pump reaches to the bottom of the tank and delivers the grit/water mixture to the separately mounted Screw Classifier. Excess water and classified solids are returned to the grit removal chamber. Grit scouring and the airlift pump are controlled automatically from the blower unit through a three-way valve arrangement. Grit removal systems supplied in the past can be updated by installing the new Jeta mechanism.



Typical Jeta® grit removal unit with a Screw Classifier. Grit extraction is with a grit pump.

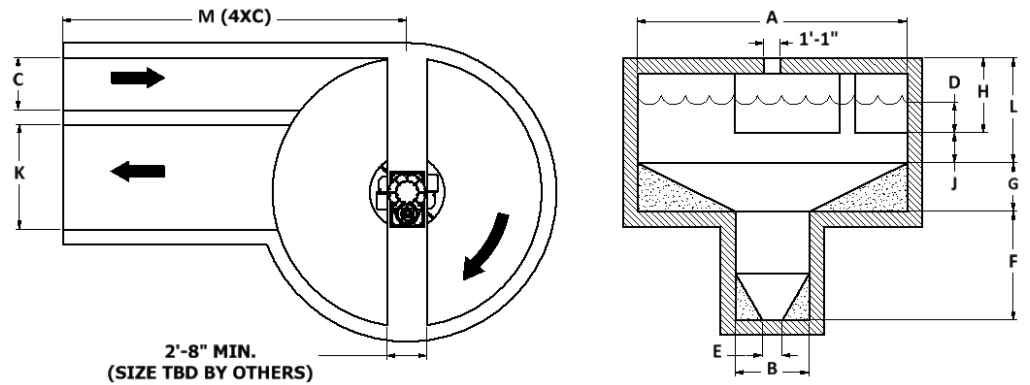
Standard design allows for the use in construction of the Jeta® chamber of pre-cast concrete rings & site formed civil structure.



Complete Jeta® grit removal systems can be supplied in steel fabricated tanks. These are ideal for attachment to existing flow channels. Jeta® grit trap sizes up to the 550 size can be supplied in this form.

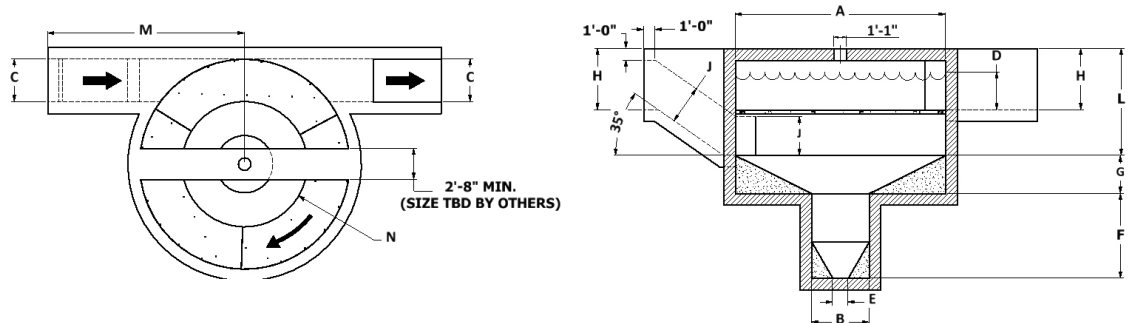
Each of the 270 and 360 design grit traps can be supplied with an annulus baffle system for improving capture.

JETA® Grit Trap 270 Dimensions



270 Type		Dia	Low Dia	Inlet Width	W.L.	Cone Bottom	Hopper Height	Slope Height	Channel Depth	I.B. to Slope	Outlet Width	Chamber Height	Channel Length 4xC
MODEL	MDG	A	B	C	D	E	F	G	H	J	K	L	M
50	1	6'	3' 4"	1' 0"	8"	1' 0"	4' 7"	1' 0"	2' 7"	1' 0"	2' 0"	3' 7"	4' 0"
100	2.5	7'	3' 4"	1' 3"	1' 0"	1' 0"	4' 7"	1' 0"	2' 7"	1' 0"	2' 6"	3' 7"	5' 0"
200	4	8'	3' 4"	1' 6"	1' 4"	1' 0"	4' 9"	1' 4"	3' 0"	1' 0"	3' 0"	4' 0"	6' 0"
300	7	10'	3' 4"	2' 0"	1' 6"	1' 0"	4' 11"	1' 6"	3' 5"	1' 0"	4' 0"	4' 5"	8' 0"
550	12	12'	4' 11"	2' 6"	1' 11"	1' 4"	5' 7"	2' 0"	3' 7"	1' 8"	5' 0"	5' 3"	10' 0"
900	20	16'	4' 11"	3' 3"	2' 6"	1' 4"	7' 3"	3' 3"	4' 4"	1' 8"	6' 6"	6' 1"	13' 0"
1300	30	18'	4' 11"	3' 6"	3' 4"	1' 4"	7' 3"	3' 3"	5' 0"	2' 0"	7' 0"	7' 0"	14' 0"
1750	40	19'	4' 11"	4' 0"	3' 10"	1' 4"	8' 3"	4' 3"	5' 6"	2' 6"	8' 0"	8' 1"	16' 0"
2000	50	20'	4' 11"	4' 6"	4' 4"	1' 4"	8' 3"	4' 3"	6' 4"	2' 6"	9' 0"	8' 10"	18' 0"
2500	70	24'	4' 11"	5' 6"	5' 6"	1' 4"	8' 3"	4' 3"	7' 6"	2' 6"	11' 0"	10' 0"	22' 0"
3500	100	28'	4' 11"	6' 7"	6' 6"	1' 4"	9' 6"	5' 0"	8' 6"	2' 6"	13' 0"	11' 0"	26' 4"

JETA® Grit Trap 360 Dimensions



360 Type		Dia	Low Dia	Inlet Width	W.L.	Cone Bottom	Hopper Height	Slope Height	Channel Depth	*270 I.B. to Slope Inlet 360 Height*	Out Width	Chamber Height	Channel Inlet Length	Inner Annulus dia
MODEL	MDG	A	B	C	D	E	F	G	H	J	K	L	M	N
50	1	6'	3' 4"	1' 0"	11"	1' 0"	4' 7"	1' 0"	2' 8"	1' 0"	NA	4' 0"	7' 7"	4' 0"
100	2.5	7'	3' 4"	1' 4"	1' 4"	1' 0"	4' 7"	1' 0"	3' 0"	1' 4"		4' 9"	8' 6"	4' 8"
200	4	8'	3' 4"	1' 8"	1' 8"	1' 0"	4' 9"	1' 4"	3' 5"	1' 8"		5' 5"	9' 5"	5' 2"
300	7	10'	3' 4"	2' 0"	2' 0"	1' 0"	4' 11"	1' 6"	3' 9"	2' 0"		6' 6"	11' 4"	6' 0"
550	12	12'	4' 11"	2' 8"	2' 3"	1' 4"	5' 7"	2' 0"	4' 2"	2' 8"		7' 5"	12' 6"	6' 8"
900	20	16'	4' 11"	3' 4"	2' 6"	1' 4"	7' 3"	3' 3"	4' 5"	3' 0"		8' 2"	15' 0"	9' 5"
1300	30	18'	4' 11"	3' 8"	3' 2"	1' 4"	7' 3"	3' 3"	5' 3"	3' 4"		9' 2"	16' 10"	10' 9"
1750	40	19'	4' 11"	3' 11"	3' 11"	1' 4"	8' 3"	4' 3"	6' 0"	3' 4"		9' 11"	18' 5"	11' 2"
2000	50	20'	4' 11"	3' 11"	4' 11"	1' 4"	8' 3"	4' 3"	7' 0"	3' 4"		11' 1"	20' 7"	12' 2"
2500	70	24'	4' 11"	4' 6"	6' 1"	1' 4"	8' 3"	4' 3"	8' 4"	4' 3"		13' 4"	24' 2"	14' 2"
3500	100	28'	4' 11"	6' 6"	7' 3"	1' 4"	9' 6"	5' 0"	9' 6"	5' 8"		15' 10"	26' 7"	16' 0"

TEXP EXPLOSION PROOF FAMILY



AEHHXV/AEHHXU, NEMA PREMIUM [XP]

AEHHXG/AEHHXF, NEMA PREMIUM, FOOTED C-FACE (1 HP - 100 HP)[XP_C]

AEUHXG/AEUHXF, NEMA PREMIUM, ROUND BODY C-FACE (1 HP - 75 HP)[XPV_C]

Effective 07-08-18
Supersedes 03-24-17



APPLICATIONS:

- Grain Elevators
- Pumps
- Blowers
- Applications Where Explosive Gases are Present
- Applications Where Explosive Dusts/ Grains are Present

FEATURES:

- Output Range: 1 - 400 HP
- Speed: 3600, 1800, 1200 & 900 RPM
- Enclosure: Totally Enclosed Fan Cooled - Explosion Proof (IP55)
- Voltage: 230/460V (Usable on 208V); 150HP and Larger is 460V Only⁽¹⁾
- Three Phase, 60 Hz, 1.15 Service Factor (Continuous); 50 Hz, 1.0 Service Factor (Continuous)
- Class I, Div. 1, Group C & D and Class II, Groups E, F and G - Temp Code T2D/T3B Up to and Including F# 256T
- Class I, Div. 1, Group D and Class II, Groups E, F and G - Temp Code T2D/T3B for F# 284T and Larger
- UL File: E84757
- CSA File: #64671
- Class F Insulation
- Class B Temperature Rise
- NEMA Design B Torques
- Standard with Klaxon 9700K Temperature Limiting Switch, 1 per Phase
- Grounding Terminal Inside Main Conduit Box
- Oversized Main Conduit Box Rotatable in 90 Degree Increments - F1 Mounted
- Dual Drilled Back Feet on 447/9 and 5007/9 Frames
- Designed for 40°C Ambient Temperature⁽²⁾
- Designed for 3300 ft. Elevation⁽³⁾
- Bi-Directional Rotation
- Cast Iron Frame, Fan Cover, End Brackets and Main Conduit Box
- Capable of Withstanding Explosion Force as Required by UL
- Stainless Steel Breather Drains with Bronze Filters (For Horizontal Mount Only)
- 1045 Carbon Steel Shaft
- Aluminum Die Cast Squirrel Cage Rotor Construction through 449 Frame; Fabricated Copper Bar on 5000 Frame
- Aluminum Die Cast Squirrel Cage Rotor Construction
- Paint System: Phenolic Rust Proof Base Plus Polyurethane Top Coat
- Paint Color: Dark Blue - Munsell 5 PB 4.5/2
- Double Shielded Bearings Pre-Packed with MULTEMP SRL for F# 140T - 280T (Non-regreasable)
- High Quality Ball (or Roller) Bearings Regreasable with Mobil Polyrex™ EM for F# 280TS and Larger
- Bronze Flinger on Both Ends
- Cast Iron Inner and Outer Bearing Caps for F#280TS and Larger
- Stainless Steel Nameplate
- Suitable for Inverter Use per NEMA MG-1 Part 31.4.4.2^(4,5)
- UL Listed for Inverter Duty^(4,5)
- Inverter Duty Magnet Wire Capable of Withstanding Voltage Spikes of up to 2200 Volts
- Inverter Duty Speed Ranges: VT = 3 - 60 Hz CT for 140T - 210T Frames = 10 - 60 Hz
CT for 250T - 320T Frames = 13 - 60 Hz CT for 260T - 440T Frames = 16 - 60 Hz
- 9 Leads for 5 HP and Smaller; 12 Leads for 7.5 HP to 125 HP; 6 Leads for 150 HP and Larger

EXTRAS/ OPTIONS:

Please refer to pages 147 - 154 for common modifications that can be performed.

Notes:

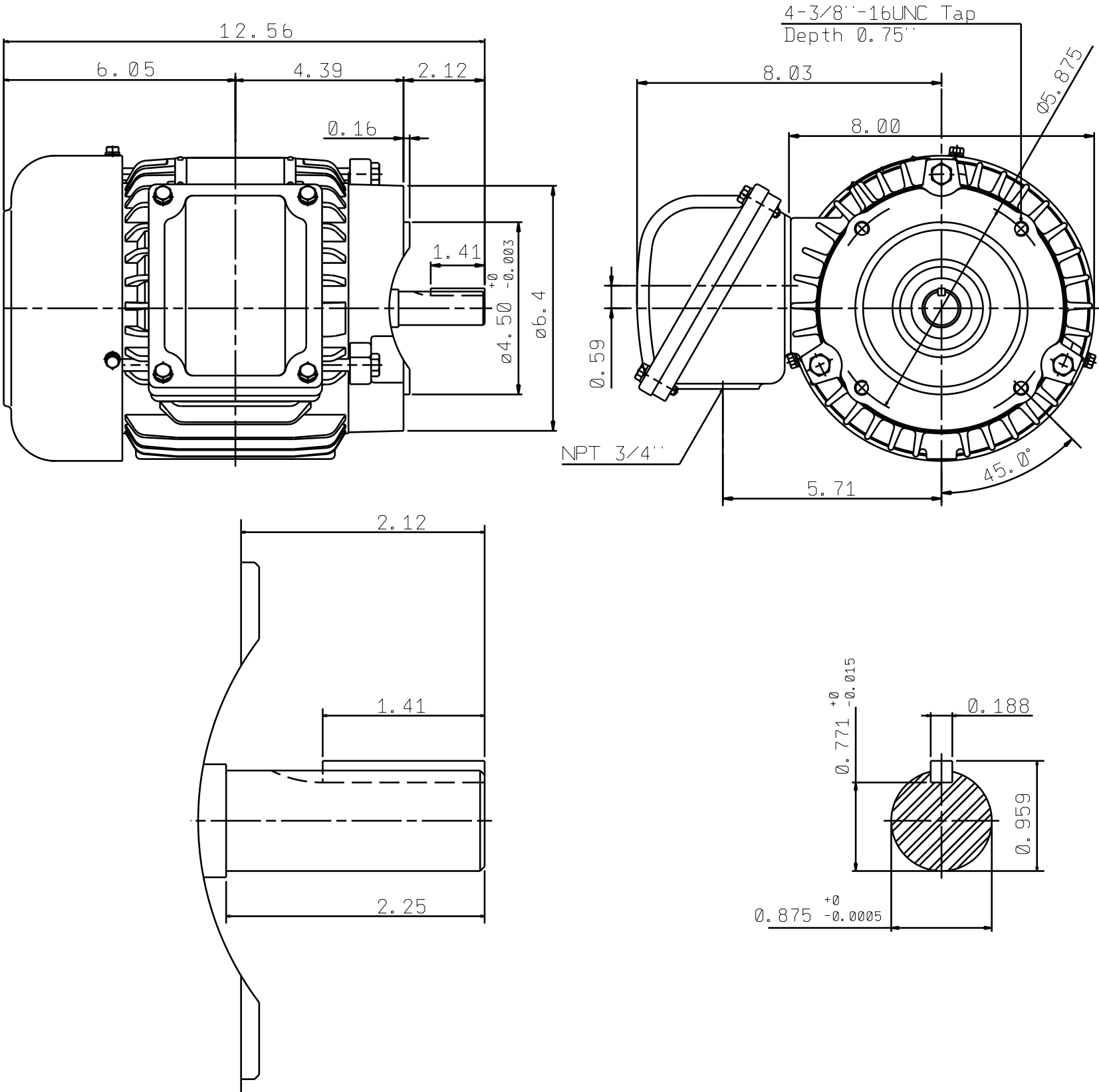
- (1) Available in 575V. TWMC carries minimal 575V stock; please check availability to ensure required motors are in stock.
Ratings may be available from our Canadian warehouses and/or our factory. Pricing and leadtime may vary.
- (2) Consult a Stock Product Application Specialist for suitability in higher ambient environments.
- (3) Consult a Stock Product Application Specialist for suitability at higher elevations.
- (4) Motor Service Factor is 1.0 when operated on a VFD.
- (5) Precautions should be taken to eliminate or reduce shaft currents that may be imposed on the motor by the VFD as stated per NEMA MG-1. Part 31.
- (6) Extras/Modification options are limited on TEXP Products. See page 147 for additional details.

DATE MAY 11. 2015	OUTLINE DIMENSIONS 3-PHASE INDUCTION MOTOR	MOTOR TYPE: AEUHXG
CATALOG NO. XPV0014C		FRAME NO. 143TC

Pole	HP	KW	Hz	VOLT	Syn. Speed RPM
4	1	.75	60	230/460	1800

Ins	Rating	Dimension in	Approx Weight	Bearings	
F	CONT.	inches	90 lbs.	DE: 6205ZZ	NDE: 6205ZZ

Totally Enclosed Fan-Cooled Type. Squirrel-Cage Rotor.



NOTE:
1.EXPLOSION PROOF:
CLASS I, DIV 1, GROUP C+D,
CLASS II, DIV 1 and 2, GROUP
E,F,G.

--	--	--	--

DWN.	J. H. LIANG	11.12.98	TEC Westinghouse	DWG NO.
CHKD.	C. S. LO	11.23.98		
APPD.	Y. B. HUANG	11.23.98		

TECO Westinghouse

ISSUED September 5, 2017	PERFORMANCE DATA 3-PHASE INDUCTION MOTOR	ENCLOSURE EXPLOSION PROOF
TYPE AEUHXG		CATALOG# XPV0014C

NAMEPLATE INFORMATION

OUTPUT		POLE	FRAME SIZE	VOLTAGE	HZ	RATED AMBIENT	INS. CLASS	NEMA DESIGN	TIME RATING	SERVICE FACTOR
HP	KW									
1	0.7	4	143T	230/460	60	40°C	F	B	CONT.	1.15

TYPICAL PERFORMANCE

FULL LOAD RPM	EFFICIENCY				POWER FACTOR			MAXIMUM POWER FACTOR CORRECTION
	FULL LOAD		3/4 LOAD %	1/2 LOAD %	F. L. %	3/4 LOAD %	1/2 LOAD %	
	MIN. %	NOM. %						
1755	82.5	85.5	84	80.5	68	58.5	45	0.9 KVAR

CURRENTS

NO LOAD			FULL LOAD			LOCKED ROTOR			NEMA KVA CODE LETTER
AT	AT	AT	AT	AT	AT	AT	AT	AT	
208 VOLT	230 VOLT	460 VOLT	208 VOLT	230 VOLT	460 VOLT	208 VOLT	230 VOLT	460 VOLT	
2.52	2.28	1.14	3.56	3.22	1.61	33.17	30.00	15.00	N

TORQUE

INERTIA

ACCEL TIME

FULL LOAD lb-ft	LOCKED ROTOR %FLT	PULL UP %FLT	BREAK DOWN %FLT	ROTOR WR ² lb-ft ²	NEMA LOAD WK ² lb-ft ²	MAX ALLOWABLE WK ² lb-ft ²	NEMA LOAD WK ² Sec	MAX ALLOWABLE WK ² Sec
2.992	350	325	450	0.075	5.8	25.31	3.12	13.49

SAFE STALL TIME IN SECONDS

ALLOWABLE STARTS PER HOUR

SOUND PRESSURE LEVEL @ 3 FT dB(A)

COLD	HOT	COLD	HOT	53
40	28	2	1	

APPROVED:

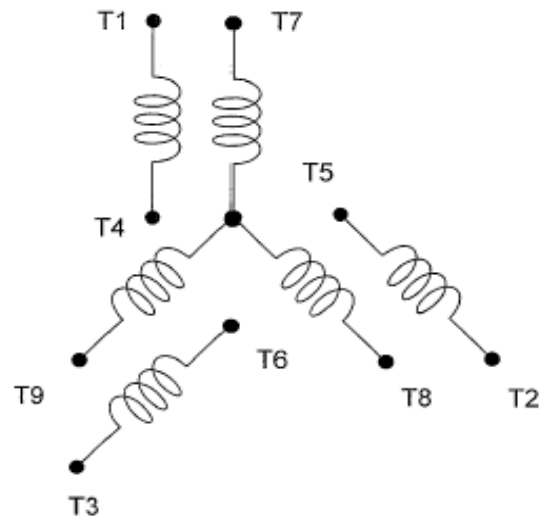
M. PRATER

DRAWING NO.

31057XPV0014C

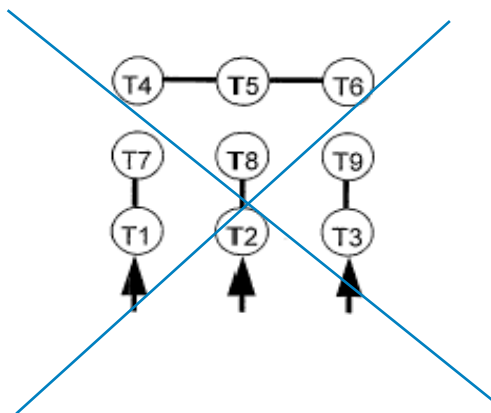
REVISION **0**

DATE: June 28, 2005	CONNECTION DIAGRAM	CATALOG NO.: XV0014C
------------------------	---------------------------	-------------------------

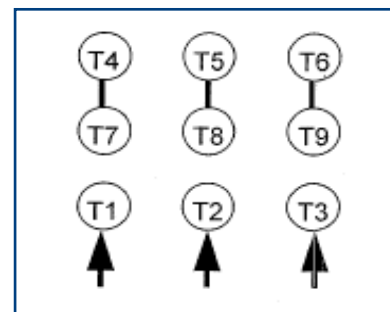


SCHEMATIC - 2Y/Y CONNECTION

ACROSS THE LINE CONNECTION

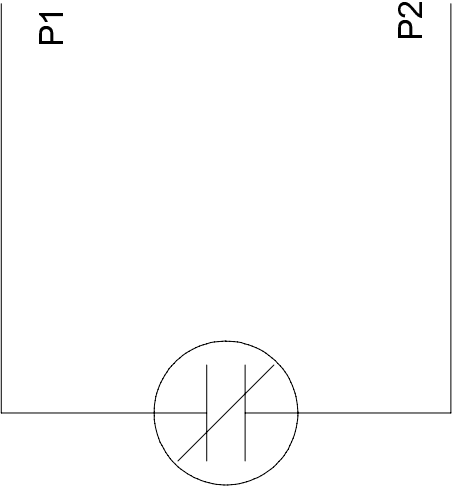


**LINE
230 VOLT CONNECTION**



**LINE
460 VOLT CONNECTION**

NORMALLY CLOSED
OPENS ON RISING TEMP



STATOR WINDING THERMOSTATS SCHEMATIC DIAGRAM

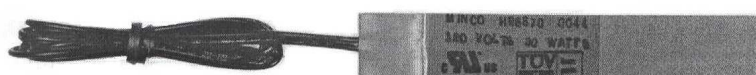
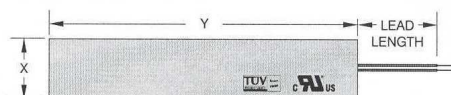
REV DATA	DFTM D. BRYAN		061581	S. O. 1625AA	REV
	APPD G. GODWIN		070881	D SPEC 110A75	02
	APPD J. BROWN	071481	APPD CULBERTSON	DIMENSIONS IN INCHES - NTS	
	APPD		APPD D.F. KOSAR	8629A28	

AMT; DEM; 04/30/99

TECO-WESTINGHOUSE MOTOR COMPANY

ROUND ROCK, TX USA

Space Heaters



These heaters have a wire element instead of a foil. They are thicker and have reduced watt densities in larger frame sizes.

- Uniform heating 220°C (428°F). Not to exceed 165°C (329°F) when installed on motor frame.
- Withstand repeated flexing
- TUV and UL component recognition marking are standard
- 2.5,5,10,15 watts/in² at 120 or 240 V. Usable on 50/60 Hz
- Lower leakage current due to capacitive coupling

General Specification data:

Temperature Range: -45 to 220°C

Thickness : 0.055 inches ±0.005

Max. voltage rating: UL and TUV recognition up to 250 Vac

Lead wires: AWG 20, PTFE insulated per UL 119/CSA. Length on standard models is 12 inches. Current capacity (based on 100°C max. ambient temp.)

AWG 20 13.5 A

Wattage tolerance: ±5%

<i>Size (inches)</i>		<i>Volts</i>	<i>Watts</i>	<i>Resistance</i>
<u>X</u>	<u>Y</u>			
1	6	120	30	480
1	10	120	50	288
1	12	120	60	240
1	20	120	100	144
1	30	120	150	96
<i>Size (inches)</i>		<i>Volts</i>	<i>Watts</i>	<i>Resistance</i>
<u>X</u>	<u>Y</u>			
1	10	240	50	1152
1	12	240	60	960
2	20	240	200	288



INSTALLATION AND MAINTENANCE INSTRUCTIONS FOR THREE PHASE INDUCTION MOTORS

Frames 56 and 143T - 449TZ



5100 North IH 35 Round Rock, Texas 78681

RECEIVING

1. Check nameplate data.
2. Check whether any damage has occurred during transportation.
3. After removal of shaft clamp, turn shaft by hand to check that it turns freely.
4. If motor is to be reshipped (alone or installed to another piece of equipment) the shaft must again be clamped to prevent axial movement.

Note: Remove the bearing clamp before turning the shaft on 284T-449TZ frame motors.

WARNING

THE FOLLOWING SAFETY PRECAUTIONS MUST BE OBSERVED:

1. Electric rotating machinery and high voltage can cause serious or fatal injury if improperly installed, operated or maintained. Responsible personnel should be familiarized with NEMA MG-1; Safety Standards for Construction and Guide Selection. Installation and Use of Electric Motors and Generators; National Electric Code and all local safety requirements.
2. When servicing, all power sources to the motor and to the accessory devices should be de-energized and disconnected and all rotating parts should be at standstill.
3. Lifting means, when supplied, are intended for lifting the motor only. When two lifting devices are supplied with the motor a dual chain must be used.
4. Suitable protection must be used when working near machinery with high noise levels.
5. Safeguard or protective devices must not be by-passed or rendered inoperative.
6. The frame of this machine must be grounded in accordance with the National Electric Code and applicable local codes.
7. A suitable enclosure should be provided to prevent access to the motor by other than authorized personnel. Extra caution should be observed around motors that are automatically or have automatic re-setting relays as they may restart unexpectedly.
8. Shaft key must be fully captive or removed before motor is started.
9. Provide proper safeguards for personnel against possible failure of motor-mounted brake, particularly on applications involving overhauling loads.
10. Explosion proof motors are constructed to comply with the label service procedure manual, repair of these motors must be made by TECO-Westinghouse Motor Company or U/L listed service center in order to maintain U/L listing.

LOCATION

1. Drip-proof motors are intended for use where atmosphere is relatively clean, dry, well ventilated and non-corrosive.
2. Totally enclosed motors may be installed where dirt, moisture, or dust are present and in outdoor locations.
3. Explosion-proof motors are built for use in hazardous locations as indicated by Underwriters' label on the motor.
4. Chemical duty enclosed motors are designed for installation in high corrosion or excessive moisture locations.

Note: in all cases, no surrounding structure should obstruct normal flow or ventilating air through or over the motor.

MOUNTING

1. Mount motor securely on a firm, flat base. All ball bearing normal thrust motors up to and including 256T frame size may be side-wall or ceiling mounted; all others check nearest TECO-Westinghouse office for mounting recommendations.
2. Align motor accurately, using a flexible coupling if possible. For drive recommendations, consult with drive or equipment manufacturer, or TECO-Westinghouse.
3. Mounting bolts must be carefully tightened to prevent changes in alignment and possible damage to the equipment. The recommended tightening torque's for medium carbon steel bolts, identified by three radial lines at 120 degrees on the head, are:

Bolt Size	Recommended Torque (Ft-lb.)	
	Minimum	Maximum
3/8	25	37
1/2	60	90
5/8	120	180
3/4	210	320

4. V-belts Sheave Pitch Diameters should not be less than those shown in Table 1 (NEMA recommended values)
5. Tighten belts only enough to prevent slippage. Belt speed should not exceed 5000 ft. per min.

TABLE 1. V-Belt Sheave Pitch Diameters (MG1-14.42)

Frame Number					V-Belt Sheave			
					Conventional A, B, C, D AND E		Narrow 3V, 5V, AND 8V	
					Minimum Pitch Diameter Inches	*Maximum Width Inches	Minimum Outside Diameter Inches	**Maximum Width Inches
	Horsepower at							
	Synchronous Speed, RPM							
3600	1800	1200	900					
143T	1.5	1	.75	.5	2.2	4.25	2.2	2.25
145T	2-3	1.5-2	1	.75	2.4	4.25	2.4	2.25
182T	3	3	1.5	1	2.4	5.25	2.4	2.75
182T	5	...	...	...	2.6	5.25	2.4	2.75
184T	...	...	2	1.5	2.4	5.25	2.4	2.75
184T	5	...	...	...	2.6	5.25	2.4	2.75
184T	7.5	5	...	...	3.0	5.25	3.0	2.75
213T	7.5-10	7.5	3	2	3.0	6.5	3.0	3.375
215T	10	...	5	3	3.0	6.5	3.0	3.375
215T	15	10	...	...	3.8	6.5	3.8	3.375
254T	15	...	7.5	5	3.8	7.75	3.8	4
254T	20	15	...	...	4.4	7.75	4.4	4
256T	20-25	...	10	7.5	4.4	7.75	4.4	4
256T	...	20	...	...	4.6	7.75	4.4	4
284T	...	...	15	10	4.6	9	4.4	4.625
284T	...	25	...	...	5.0	9	4.4	4.625
286T	...	30	20	15	5.4	9	5.2	4.625

TABLE 1. V-Belt Sheave Pitch Diameters (MG1-14.42)

Frame Number					V-Belt Sheave			
					Conventional A, B, C, D AND E		Narrow 3V, 5V, AND 8V	
					Minimum Pitch Diameter Inches	*Maximum Width Inches	Minimum Outside Diameter Inches	**Maximum Width Inches
	Horsepower at Synchronous Speed, RPM							
	3600	1800	1200	900				
324T	...	40	25	20	6.0	10.25	6.0	5.25
326T	...	50	30	25	6.8	10.25	6.8	5.25
364T	...	...	40	30	6.8	11.5	6.8	5
364T	...	60	...	...	7.4	11.5	7.4	5.785
365T	...	...	50	40	8.2	11.5	8.2	5.785
365T	...	75	...	...	9.0	11.5	8.6	5.785
404T	...	...	60	...	9.0	14.25	8.0	7.25
404T	...	...	...	50	9.0	14.25	8.4	7.25
404T	...	100	...	...	10.0	14.25	8.6	7.25
405T	...	...	75	60	10.0	14.25	10.0	7.25
405T	...	100	...	...	10.0	14.25	8.6	7.25
405T	...	125	...	...	11.5	14.25	10.5	7.25
444T	...	...	100	...	11.0	16.75	10.0	8.5
444T	...	...	...	75	10.5	16.75	9.5	8.5
444T	...	125	...	...	11.0	16.75	9.5	8.5
444T	...	150	...	...	...	16.75	10.5	8.5
445T	...	...	125	...	12.5	16.75	12.0	8.5
445T	...	...	...	100	12.5	16.75	12.0	8.5
445T	...	150	...	...	...	16.75	10.5	8.5

*Max. Sheave width = 2(N-W) - .25

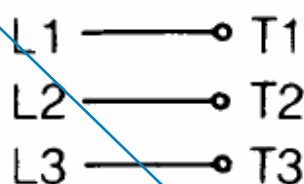
**Max Sheave width = N-W

***Sheave ratios greater than 5:1 and center-to-center distance less than the diameter of the large sheave should be referred to TECO-Westinghouse.

POWER SUPPLY & CONNECTIONS

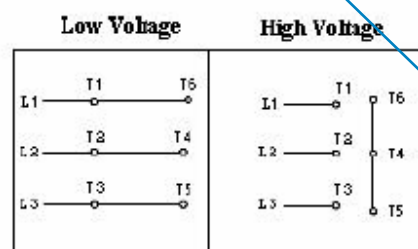
1. Wiring of motor and control, overload protection and grounding should be in accordance with National Electrical Code and all local safety requirements.
2. Nameplate voltage and frequency should agree with power supply. Motor will operate satisfactorily on line voltage within $\pm 10\%$ of nameplate voltage; or frequency with $\pm 5\%$ and with a combined variation not to exceed $\pm 10\%$. 230-volt motors can be used on 208-volt network systems, but with slightly modified performance characteristics as shown on the nameplate.
3. Dual voltage and single voltage motors can be connected for the desired voltage by following connection diagram shown on the nameplate or inside of the conduit box.
4. All Explosion Proof motors have Temperature Limiting Devices in the motor enclosure to prevent excessive external surface temperature of the motor in accordance with U/L standards. Terminals of thermal protectors (P1 & P2) should be connected to the motor control equipment, according to the connection diagram inside of the conduit box.
5. Standard connection diagram for three phase, not thermally protected, dual rotation motors are shown in diagrams A through E. **(Note: To change rotation, Interchange any two line leads)**

A. 3 Lead, Single Voltage

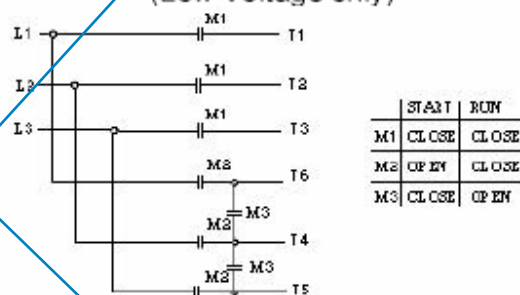


B. 6 Lead, Dual Voltage & Voltage Ratio 1 to 3

B-1 Across the Line Start & Run

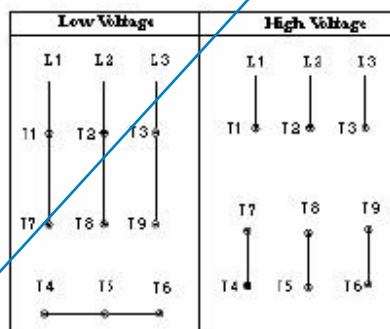


B-2 Wye Start & Delta Run
(Low Voltage only)

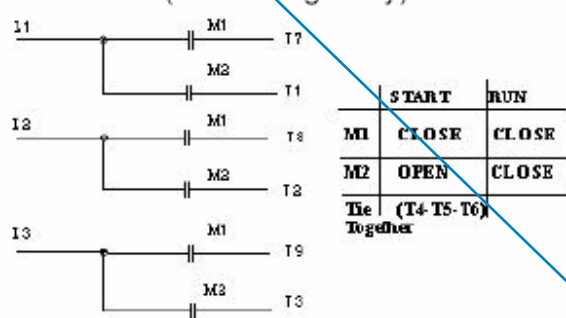


C. 9 Leads; Dual Voltage & Voltage Ratio 1 to 2, Wye Connected

C-1 Across the Line Start & Run

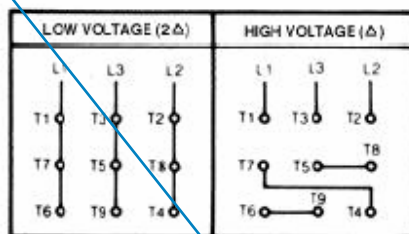


C-2 Part Winding Start
(Low Voltage only)

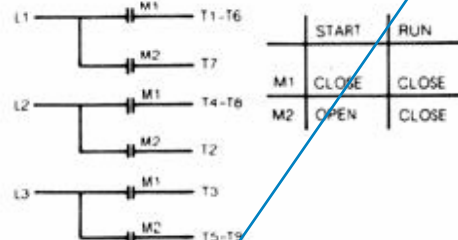


D. 9 Leads; Dual Voltage & Voltage Ratio 1 to 2, Delta Connected

D-1 Across the Line Start & Run

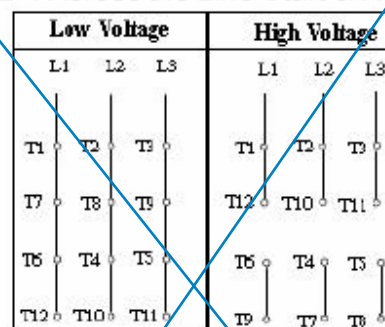


D-2 Part Winding Start (Low Voltage only)

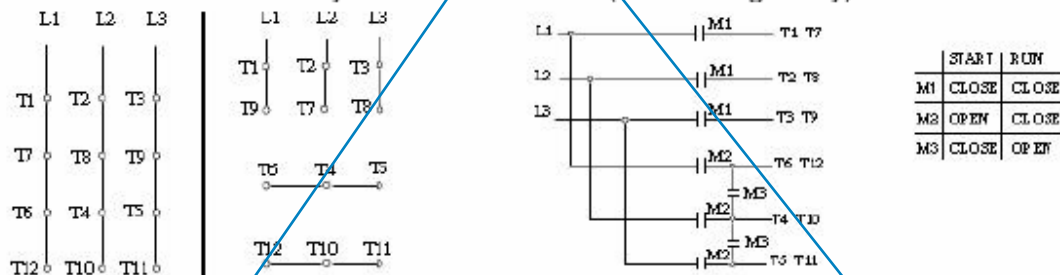


E. 12 Leads, Dual Voltage

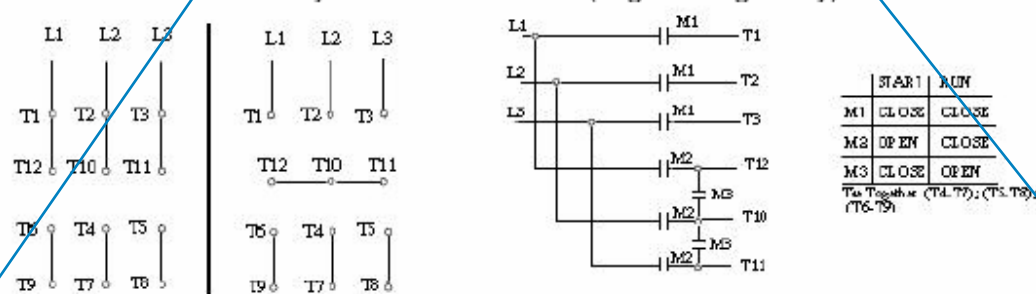
E-1 Across the Line Start & Run



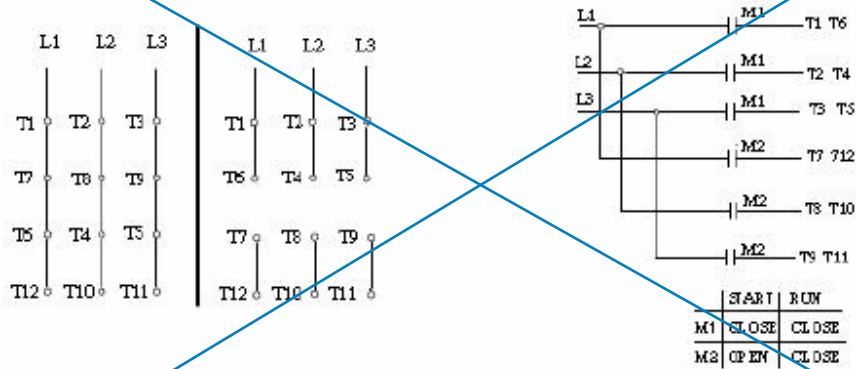
E-2-1 Wye Start & Delta Run (Low Voltage only)



E-2-2 Wye Start & Delta Run (High Voltage only)



E-3 Part Winding Start (Low Voltage only)



*Important: For Part Winding Start, M2 contactor should be closed within two (2) seconds after M1 contactor is closed.
Only 4 pole and above (e.g., 6P, 8P...) motors are satisfactory for Part Winding Start at low voltage.

START UP

1. Disconnect load and start motor. Check direction of rotation. If rotation must be changed, ALLOW THE MOTOR TO STOP COMPLETELY. Interchange any two leads of a three-phase motor.
2. Connect load. The motor should start quickly and run smoothly. If no, shut power off at once. Recheck the assembly including all connections before restarting.
3. If excessive vibration is noted, check for loose mounting bolts too flexible motor support structure or transmitted vibration from adjacent machinery. Periodic vibration checks should be made; foundations often settle.
4. Operate under load for short period of time and check operating current against nameplate.

TESTING

If the motor has been in storage for an extensive period or has been subjected to adverse moisture conditions, it is best to check the insulation resistance of the stator winding with a megometer. Depending on the length and conditions of storage it may be necessary to regrease or change rusted bearings.

If the resistance is lower than one megohm the windings should be dried in one of the following two ways:

1. Bake in oven at temperatures not exceeding 194°F until insulation resistance becomes constant.
2. With rotor locked, apply low voltage and gradually increase the current through windings until temperature measured with a thermometer reaches 194°F. Do not exceed this temperature.

MAINTENANCE

INSPECTION

Inspect motor at regular intervals. Keep motor clean and ventilation openings clear.

LUBRICATION

1. Frame 143T-256T: Double shielded and pre-lubricated ball-bearing motors without grease fittings and don't need re-lubrication, except on MAX-E1[®] and MAX-E2[®] products which have re-greasable features.
2. Frames 280TS, 320-449TZ(TS): Motors having grease fittings and grease discharge devices at brackets. Motors are shipped with grease for initial running. It is necessary to re-lubricate anti-friction bearing motors periodically, depending on size and type of service. See Table 2 to provide maximum bearing life. Excessive or too frequent lubrication may damage the motor.

TABLE 2

Horsepower	Standard Conditions	Severe Conditions	Extreme Conditions
1 Thru 30 Hp, 1800 rpm and below	7 years	3 years	180 days
40 Thru 75 Hp, 1800 rpm and below	210 days	70 days	30 days
100 Thru 150 Hp, 1800 rpm and below	90 days	30 days	15 days
1 Thru 20 Hp, 3600 rpm	5 years	2 years	90 days
25 Thru 75 Hp, 3600 rpm	180 days	60 days	30 days
100 Thru 150 Hp, 3600 rpm	90 days	30 days	15 days

Note:

- A. Standard conditions: 8 hours operation per day, normal or light loading, clear and 40°C ambient conditions.
 - B. Severe conditions: 24-hour operation per day or light shock loading, vibration or in dirty or dusty conditions.
 - C. Extreme conditions: With heavy shock loading or vibration or dusty conditions.
 - D. For double shielded bearings, above data (lubrication frequency) means that the bearing must be replaced.
3. Be sure fittings are clean and free from dirt. Using a low-pressure grease gun, pump in the recommended grease until new grease appears at grease discharge hole.
 4. Use the POLYUREA grease unless special grease is specified on the nameplate.
 5. If re-lubrication is to be performed with the motor running, stay clear of rotating parts. After re-greasing, allow the motor to run for ten to thirty minutes.

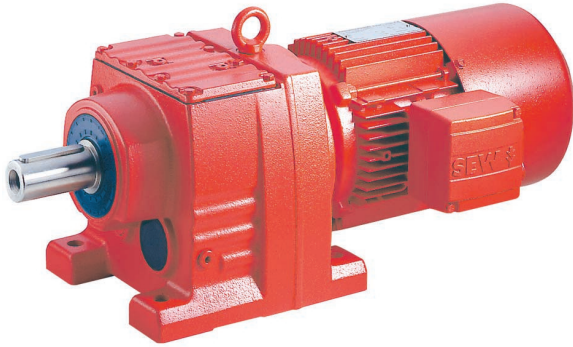
RENEWAL PARTS

1. Use only genuine TECO-Westinghouse renewal parts or as recommended by TECO-Westinghouse Motor Company.
2. When you order renewal parts please specify complete information to TECO-Westinghouse office/agent such as type, frame no., poles, horsepower, voltage, series no., quantity, etc.

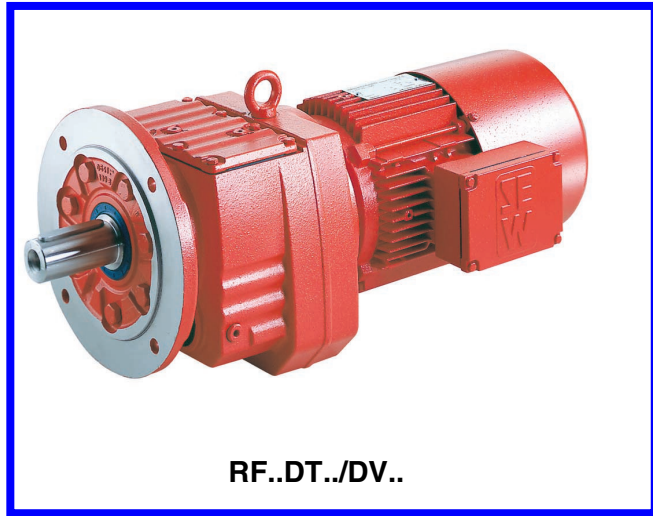
**FOR FURTHER INFORMATION PLEASE CONTACT
TECO-WESTINGHOUSE MOTOR COMPANY**

Round Rock, TX

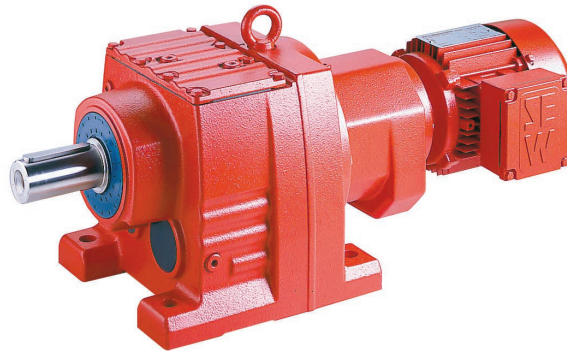
800-873-8326



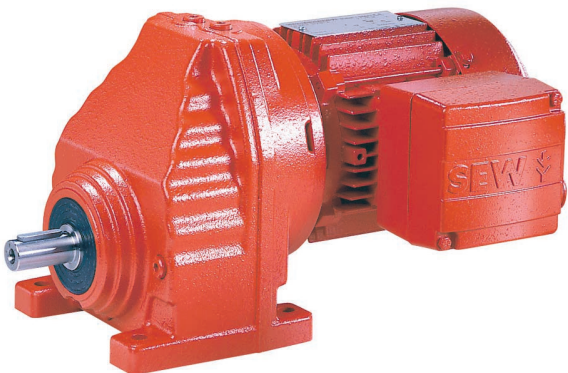
R..DT../DV..BM(G)



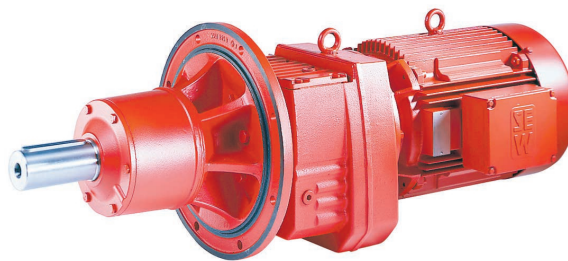
RF..DT../DV..



R..R..DT../DV..

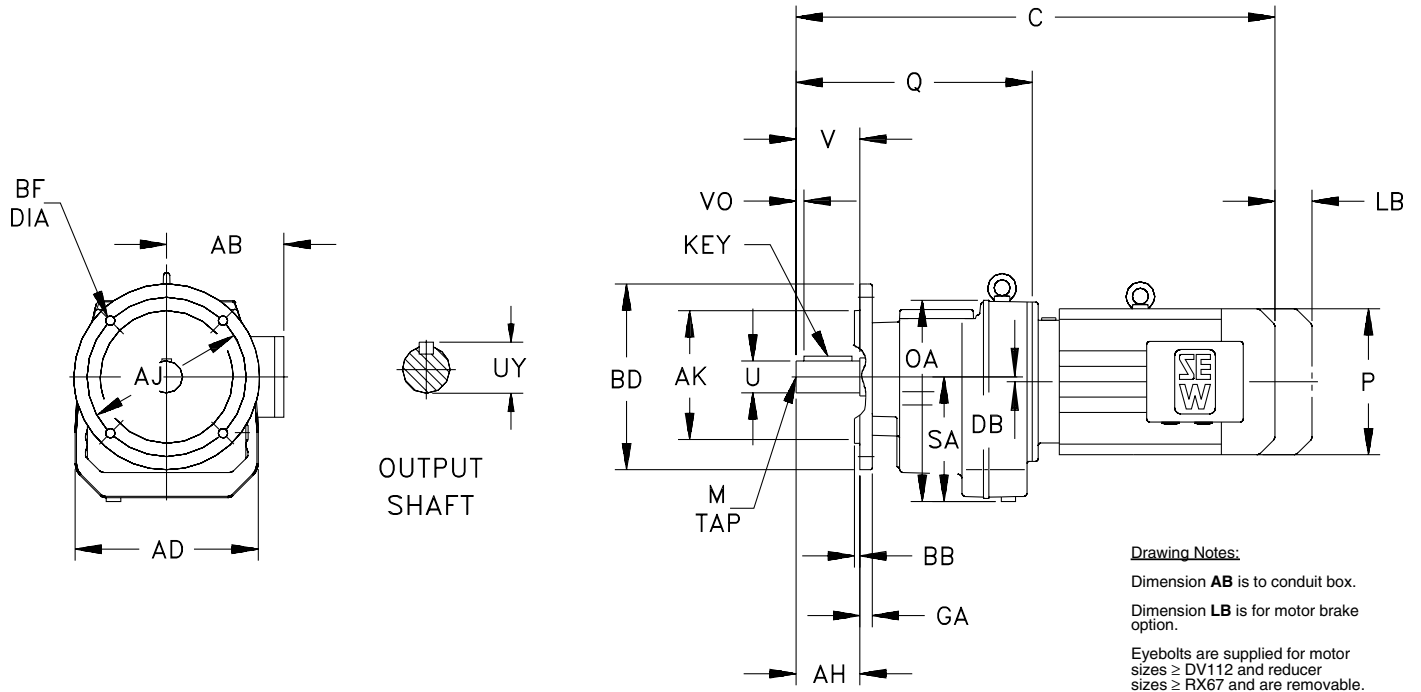


RX..DT../DV..



RM..DT../DV..

Dimensions Type RF Gearmotors - Flange Mounted



Drawing Notes:

Dimension **AB** is to conduit box.

Dimension **LB** is for motor brake option.

Eyebolts are supplied for motor sizes $\geq$ DV112 and reducer sizes $\geq$ RX67 and are removable.

Gearcase

Model	AD	DB	OA	Q	SA
RF77	9.25 235	0.63 15.9	9.13 232	11.81 300	5.67 144
RF87	11.69 297	0.50 12.6	11.77 299	14.65 372	7.24 184

Output Shaft

Inch Series/Optional Metric Series

U*	UY	V	VO	Key	M
1.625 40	1.79 43	3.15 80	0.38 5	$\frac{3}{8} \times \frac{3}{8} \times 2\frac{1}{4}$ 12 x 8 x 70	$\frac{5}{8} - 11 \times 1.38$ M16 x 36
2.125 50	2.35 53.5	3.94 100	0.64 10	$\frac{1}{2} \times \frac{1}{2} \times 2\frac{5}{8}$ 14 x 9 x 80	$\frac{3}{4} - 10 \times 1.61$ M16 x 36

* Note: See page 33 for applicable tolerances.

Flange (Specify BD dimension when ordering)

Model		AH	AJ	AK*	BB	BD	BF	GA
RF77	Option 1	3.15 80	10.43 265	9.055 230	0.16 4	11.81 300	0.53 13.5	0.73 18.5
	Option 2	3.15 80	8.46 215	7.087 180	0.16 4	9.84 250	0.53 13.5	0.73 18.5
RF87	Option 1	3.94 100	11.81 300	9.843 250	0.20 5	13.78 350	0.69 17.5	0.83 21
	Option 2	3.94 100	10.43 265	9.055 230	0.16 4	11.81 300	0.53 13.5	0.77 19.5

* Note: See page 33 for applicable tolerances.

Motor

Model		DT				DV					
		71	80	90	100	112M	132S	132M	132ML	160M	160L
	AB	5.43 138	5.43 138	6.73 171	6.89 175	7.40 188	7.40 188	9.13 232	9.13 232	10.04 255	10.55 268
	LB	2.52 64	2.52 64	3.35 85	3.35 85	3.15 80	3.15 80	4.41 112	4.41 112	4.41 112	6.14 156
	P	5.71 145	5.71 145	7.76 197	7.76 197	8.70 221	8.70 221	10.83 275	10.83 275	10.83 275	13.03 331
RF77	C	19.41 493	21.38 543	22.09 561	24.06 611	25.47 647	27.24 692	28.03 712	30.39 772	30.39 772	—
RF87	C	—	24.02 610	24.76 629	26.73 679	28.11 714	29.88 759	30.67 779	33.03 839	33.03 839	34.92 887

Dimensions are $\frac{\text{inch}}{\text{mm}}$
See page 152 for available output shaft sizes.

SEW-EURODRIVE SPECIFICATIONS



APPLICATION: JETTA DRIVE

GEAR REDUCER SPECIFICATIONS

MODEL TYPE: RF77AM143
SHAFT: 40 mm
FLANGE: 250 mm
MOUNTING POS. M4
RATIO: 52.07
RATED TORQUE: 7,260 in lb.
OUTPUT TORQUE: 1,834 in lb.
SERVICE FACTOR: 1.6

LUBRICANTS

	6)		ISO, NLGI	Mobil®	Shell	KLOBER	ARAL	BP	Tribol	TEXACO	Optimat	FUCHS
	°C	Standard										
R...												
												
K...(HK...)												
												
F...												
												
S...(HS...)												
												
R..., K...(HK...), F..., S...(HS...)												
												
W...(HW...)												
												
R32 R302												

01 805 792US

CLP	= Mineral oil	1)	Helical-worm gear units with PG oil: Please contact SEW-EURODRIVE
CLP PG	= Polyglycol (W gear units, conforms to USDA-H1)	2)	Special lubricant for Spiroplan® gear units only
CLP HC	= Synthetic hydrocarbons	3)	Recommendation: Select SEW $f_b \geq 1.2$
E	= Ester oil (water pollution danger category WGK 1)	4)	Pay attention to critical starting behavior at low temperatures!
HCE	= Synthetic hydrocarbons + ester oil (USDA - H1 certification)	5)	Low-viscosity grease
HLP	= Hydraulic oil	6)	Ambient temperature
	= Synthetic lubricant (= synthetic anti-friction bearing grease)		Lubricant for the food industry (food grade oil)
	= Mineral lubricant (= mineral-based anti-friction bearing grease)		Biodegradable oil (lubricant for use in agriculture, forestry and water resources)

Oil levels and oil quality should be checked at frequent intervals, depending on usage. Oil changes are required at intervals of 10,000 operating hours or every two years, whichever comes first. If a synthetic oil lubricant is used, then this period can be extended to 20,000 operating hours or every four years, whichever comes first. In applications where hostile operating conditions exist, such as high humidity, corrosive environment, or large temperature changes, the lubricant should be changed at more frequent intervals.

The gear units W10, W20 and W30 are supplied with a synthetic oil which is good for the life of the reducer, independent of the mounting position.

Grease packed bearings should be cleaned and regreased every 10,000 hours or 20,000 hours for synthetic grease. Input (high speed) bearings should not be overgreased. They should be filled with grease not to exceed 1/3 of the bearing's free volume. For output bearings and bearings with replaceable grease shields, fill to 2/3 of their free volume.

ATTENTION

When the recommended lubricant is not available, it is permissible to use a lubricant having equivalent characteristics but we do not recommend that lubricants of different brands be mixed. Under no circumstances should synthetic lubricants be mixed with one another or with one having a mineral base.

Mobilgear 600 XP Series

Gear Oil

Product Description

The Mobilgear 600 XP Series are extra high performance gear oils having outstanding extreme pressure characteristics and load-carrying properties, intended for use in all types of enclosed gear drives with circulation or splash lubrication systems. Mobilgear 600 XP Series is designed to stay ahead of the changing needs of gearbox technology. Gearbox technology design trends are towards smaller units with similar power throughput. This increase in power density places increased demands on gear oils. Mobilgear 600 XP Series oils are formulated to meet the stress by providing extra protection for gears, bearings and seals.

Mobilgear 600 XP Series is formulated to protect gear teeth from wear at its earliest stages. Microscopic wear, called micropitting, can lead to significant gear tooth damage. Mobilgear 600 XP Series exceeds the industry requirement for bearing wear protection. In fact, Mobilgear 600 XP Series provides up to 15 times the wear protection as measured by the industry standard FAG FE 8 test. Mobilgear 600 XP Series' balanced formulation is able to provide maximum wear and corrosion protection while maintaining compatibility with common gearbox seal materials. Mobilgear 600 XP helps to maintain gearbox seal integrity thereby preventing oil leaks and keeping contamination out. By protecting the gears, bearings and seals, Mobilgear 600 XP can improve equipment reliability and increase productivity.

Mobilgear 600 XP Series oils are recommended for industrial spur, helical and bevel enclosed gears with circulation or splash lubrication, operating at bulk oil temperatures up to 100°C. They are particularly suitable for gear sets working under heavy or shock loads. Mobilgear 600 XP oils also find broad application in marine gearing applications. They may also be used in non-gear applications include highly loaded and slow speed plain and rolling contact bearings.

Features and Benefits

Mobilgear 600 XP Series products are a leading member of the Mobil brand of industrial lubricants that enjoy a reputation for innovation and high performance capability. These mineral-based products are designed to provide high quality industrial gear oils, meeting the latest industry standards and with high versatility to lubricate a broad range of industrial and marine equipment.

Mobilgear 600 XP Series products offer the following features and potential benefits:

Features	Advantages and Potential Benefits
Enhanced gear wear protection from micropitting	Less gear and bearing wear resulting in less unexpected downtime
Reduced debris denting from generated wear particles	Up to 22% improvement in bearing life reducing bearing replacement costs and improving productivity
Improved bearing wear protection	Improved bearing life resulting in higher productivity
Outstanding compatibility with a range of seal materials	Reduced leakage, oil consumption and contamination ingress helping to reduce maintenance, extend gearbox reliability and higher productivity



Features	Advantages and Potential Benefits
Excellent resistance to oil oxidation and thermal degradation	Helps extend lubricant life with lower lubricant and lubrication costs and reduced scheduled downtime.
High resistance to sludge and deposit formation	Cleaner systems and reduced maintenance
Wide range of applications	Fewer grades of lubricant required because of wide range of application, leading to lower purchase and storage costs and less danger of using wrong lubricant
Optimised resistance to rust and corrosion of steel and corrosion of copper and soft metal alloys	Excellent protection of machine parts, with reduced maintenance and repair costs
Resistance to foaming and emulsion formation	Effective lubrication and problem free operation in the presence of water contamination or in equipment prone to oil foaming

Applications

Mobilgear 600 XP lubricants are used in a wide range of industrial and marine applications, especially spur, helical, bevel and worm gearing. Specific applications include:

- Industrial gearing for conveyers, agitators, dryers, extruders, fans, mixers, presses, pulpers, pumps (including oil well pumps), screens, extruders and other heavy duty applications
- Marine gearing including main propulsion, centrifuges, deck machinery such as winches, windlasses, cranes, turning gears, pumps, elevators and rudder carriers
- Non-gear applications include shaft couplings, screws and heavily loaded plain and rolling contact bearings operating at slow speeds.

Specifications and Approvals

Mobilgear 600 XP Series meets or exceeds the following industry specifications:	Mobilgear 600 XP 68	Mobilgear 600 XP 100	Mobilgear 600 XP 150	Mobilgear 600 XP 220	Mobilgear 600 XP 320	Mobilgear 600 XP 460	Mobilgear 600 XP 680
Flender BA 7300 Table A		Approved	Approved	Approved	Approved	Approved	Approved
AGMA 9005-E02	2 EP	3 EP	4 EP	5 EP	6 EP	7 EP	-
DIN 51517 Part 3	CLP 68	CLP 100	CLP 150	CLP 220	CLP 320	CLP 460	CLP 680
ISO 12925-1 Type	CKD 68	CKD 100	CKD 150	CKD 220	CKD 320	CKC 460	CKC 680
Textron Power Transmission Grade	E2	E3	E4	E5	E6	E7	E8
Mueller Weingarten DT 55 005		CLP 100 Approved	CLP 150 Approved	CLP 220 Approved	CLP 320 Approved	CLP 460 Approved	

Typical Properties

Mobilgear 600 XP	68	100	150	220	320	460	680
ISO Viscosity Grade	68	100	150	220	320	460	680
Viscosity, ASTM D 445							
mm ² /s @ 40°C	68	100	150	220	320	460	680
mm ² /s @ 100°C	8.8	11.2	14.7	19.0	24.1	30.6	39.2

Mobilgear 600 XP	68	100	150	220	320	460	680
Viscosity Index, ASTM D 2270	101	97	97	97	97	96	90
Pour Point, °C, ASTM D 97	-27	-24	-24	-24	-24	-15	-9
Flash Point, °C, ASTM D 92	230	230	230	240	240	240	285
Density @ 15.6 °C, ASTM D 4052, kg/l	0.88	0.88	0.89	0.89	0.90	0.90	0.91
FZG Micropitting, FVA 54, Fail Stage / Rating		10 / High	10 / High	10 / High	10 / High	10 / High	10 / High
FE 8 wear test, DIN 51819-3, D7,5/80-80. Roller wear, mg	2	2	2	2	2	2	2
Timken OK Load, ASTM D 2782, lb	65	65	65	65	65	65	65
4-Ball EP test, ASTM D 2783,							
Weld Load, kg	200	200	250	250	250	250	250
Load Wear Index, kgf	47	47	47	48	48	48	48
FZG Scuffing, Fail Stage							
A/8.3/90	12+	12+	12+	12+	12+	12+	12+
A/16.6/90		12+	12+	12+	12+	12+	12+
Rust protection, ASTM D 665, Sea Water	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Copper Strip Corrosion, ASTM D 130, 3 hrs @ 100°C	1B	1B	1B	1B	1B	1B	1B
Demulsibility, ASTM D 1401, Time to 3ml emulsion, minutes @ 82°C	30	30	30	30	30	30	30
Foam Test, ASTM D 892, Tendency/Stability, ml/ml Sequence 1	0/0	0/0	0/0	0/0	0/0	0/0	0/0
Sequence 2	30/0	30/0	30/0	30/0	30/0	30/0	30/0

Health and Safety

Based on available information, this product is not expected to produce adverse effects on health when used for the intended application, following the recommendations provided in the Material Safety Data Sheet (MSDS). MSDSs are available upon request through your sales contract office, or via the Internet on <http://www.exxonmobil.com>. This product should not be used for purposes other than its intended use. If disposing of used product, take care to protect the environment.

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ExxonMobil Lubricants & Specialties

All products may not be available locally. For more information, contact your local sales office or visit www.exxonmobil.com.

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LUBRICANTS

Dynalife® 220

Phillips 66® Dynalife 220 is a multipurpose, extreme pressure (EP), lithium grease developed for use in a wide variety of industrial and automotive applications. It is a versatile, general purpose grease for use in applications where operating temperatures and loads are moderate.

Dynalife 220 is manufactured with high quality base oils and a lithium 12-hydroxystearate soap thickener. It is fortified with EP and antiwear additives plus rust and oxidation inhibitors. It has good load carrying capacity to protect against wear and good oxidation resistance for long service life. It protects metal parts against rust and corrosion, and is resistant to water washout. It has good resistance to shearing and will not soften excessively during high speed operation in rolling element bearings maintaining a good seal to minimize bearing contamination.

Dynalife 220 is available in multiple grades for use in a wide variety of applications. It is an excellent grease for use in centralized lube systems. The semi-fluid grades may be used in enclosed industrial gearboxes where oil leakage is a problem.

Applications

- Plain and rolling-element bearings in compressors, fans, centrifugal pumps and other industrial equipment where operating temperatures and loads are moderate
- Chassis parts, universal joints and water pumps on passenger cars, light trucks and other mobile equipment
- Track rollers, commercial lawn mowers, crushers and vibrating screens (semi-fluid grades)
- Enclosed gear drives on industrial, mining and mobile equipment where oil leakage is a problem (semi-fluid grades)
- Wheel bearings of passenger cars equipped with drum brakes
- Centralized grease lubrication systems
- General purpose shop/industrial grease

Features/Benefits

- Multipurpose grease for use in a wide variety of applications
- Good EP and antiwear properties
- Excellent shear stability
- Good oxidation resistance and thermal stability
- Protects against rust and corrosion
- Excellent resistance to water washout
- Suitable for use in centralized lube systems

**Multipurpose
Extreme Pressure
Lithium Grease**

**KEEPING THE
WORLD
RUNNING
SMOOTHLY.**





Dynalife® 220

Typical Properties						
NLGI Grade		000	00	0	1	2
Thickener Type		Lithium	Lithium	Lithium	Lithium	Lithium
Color		Amber	Amber	Amber	Amber	Amber
Texture		Smooth	Smooth	Smooth	Smooth/ Tacky	Smooth/ Tacky
Density, lbs/gal		7.50	7.50	7.50	7.50	7.50
Dropping Point, °C (°F)	ASTM D2265	N/A	N/A	180 (356)	180 (356)	180 (356)
Viscosity, Kinematic	ASTM D445					
Viscosity @ 40°C, cSt		320	220	220	220	220
Viscosity @ 100°C, cSt		21.6	18.0	18.0	18.0	18.0
Penetration, Worked (60 strokes)	ASTM D217	445-475	400-430	355-385	310-340	265-295
Oxidation Stability, 100 Hours, psi (kPa)	ASTM D942	4.6 (32)	6.2 (43)	6.2 (43)	6.2 (43)	6.2 (43)
Four-Ball Wear Scar, mm	ASTM D2266	0.5	0.5	0.45	0.45	0.45
Four-Ball EP, Weld Load, Kgf	ASTM D2596	250	250	250	315	315
Timken OK Load, lbs	ASTM D2509	40	40	40	50	50
Corrosion Prevention	ASTM D1743	Pass	Pass	Pass	Pass	Pass
Copper Corrosion	ASTM D4048	1b	1b	1b	1b	1b

Health & Safety Information

For recommendations on safe handling and use of this product, please refer to the Safety Data Sheet via <http://www.phillips66.com/SDS>.

01-2019

Typical properties are average values only and do not constitute a specification. Minor variations that do not affect product performance are to be expected during normal manufacture, and at different blending locations. Product formulations are subject to change without notification.

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SUBMITTAL DATA SHEET

PROJECT NAME _____ Hilo WWTP, HI

LOCATION _____ Honolulu, HI

CUSTOMER _____ Nan, Inc.

CUSTOMER P.O. NUMBER _____ 24077-00036

OVIVO S.O. NUMBER _____ CSW0001868-02

EQUIPMENT DESCRIPTION _____ Vortex Grit Removal System

SPECIFICATION SECTION _____ 11353A

CONSULTING ENGINEER _____ Carollo

ITEM DESCRIPTION UNDER THIS COVER:

CERTIFIED DATA AND CAPABILITY..... (1) Page

**REPORT TO:**

OVIVO USA, LLC (Ovivo Water)
Attn: Jacob Thompson
Legacy @ Lake Park Blvd
4255 Lake Park Boulevard – Suite #100
Salt Lake City, UT 84120-8201

Date Sample(s) Collected: 9/14/2011
Date Received @ Lab: 10/20/2011
Date(s) Tested: 11/10/2011

Laboratory Report #: N1151676
Report Date: 11/16/2011

Email Address:

jacob.thompson@ovivowater.com

Copy (via email) to:

roger.sturt@ovivowtaer.com

<u>Sample Description:</u>	Grit
<u>Site:</u>	<u>Florence Regional – WWMF == CWS488 – 01,02</u>
<u>Influent Sample Numbers:</u>	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
<u>Effluent Sample Numbers:</u>	1, 2, 3, 4, 5, 6, 7, 8, 9, 10

<u>Sieve #:</u>	<u>Grit Reduction, Influent to Effluent</u>	<u>Units</u>
50	99.6	%
70	94.4	%
100	67.2	%

Results expressed on an ash weight (inorganic) basis, using sample average:

<u>Moisture</u>	<u>Results</u>	<u>Units</u>
Influent	62.1	%
Effluent	94.4	%

<u>Ash</u>	<u>Results</u>	<u>Units</u>
Influent	93.4	% (dry weight)
Effluent	26.7	% (dry weight)

% = percent

Comments:

- Results are based on samples as received by Northeast Laboratories, Inc. on: 10/20/2011

Approved By: _____
Laboratory Director

SUBMITTAL DATA SHEET

PROJECT NAME _____ Hilo WWTP, HI

LOCATION _____ Honolulu, HI

CUSTOMER _____ Nan, Inc.

CUSTOMER P.O. NUMBER _____ 24077-00036

OVIVO S.O. NUMBER _____ CSW0001868-02

EQUIPMENT DESCRIPTION _____ Vortex Grit Removal System

SPECIFICATION SECTION _____ ~~11353A~~ 11322D

CONSULTING ENGINEER _____ Carollo

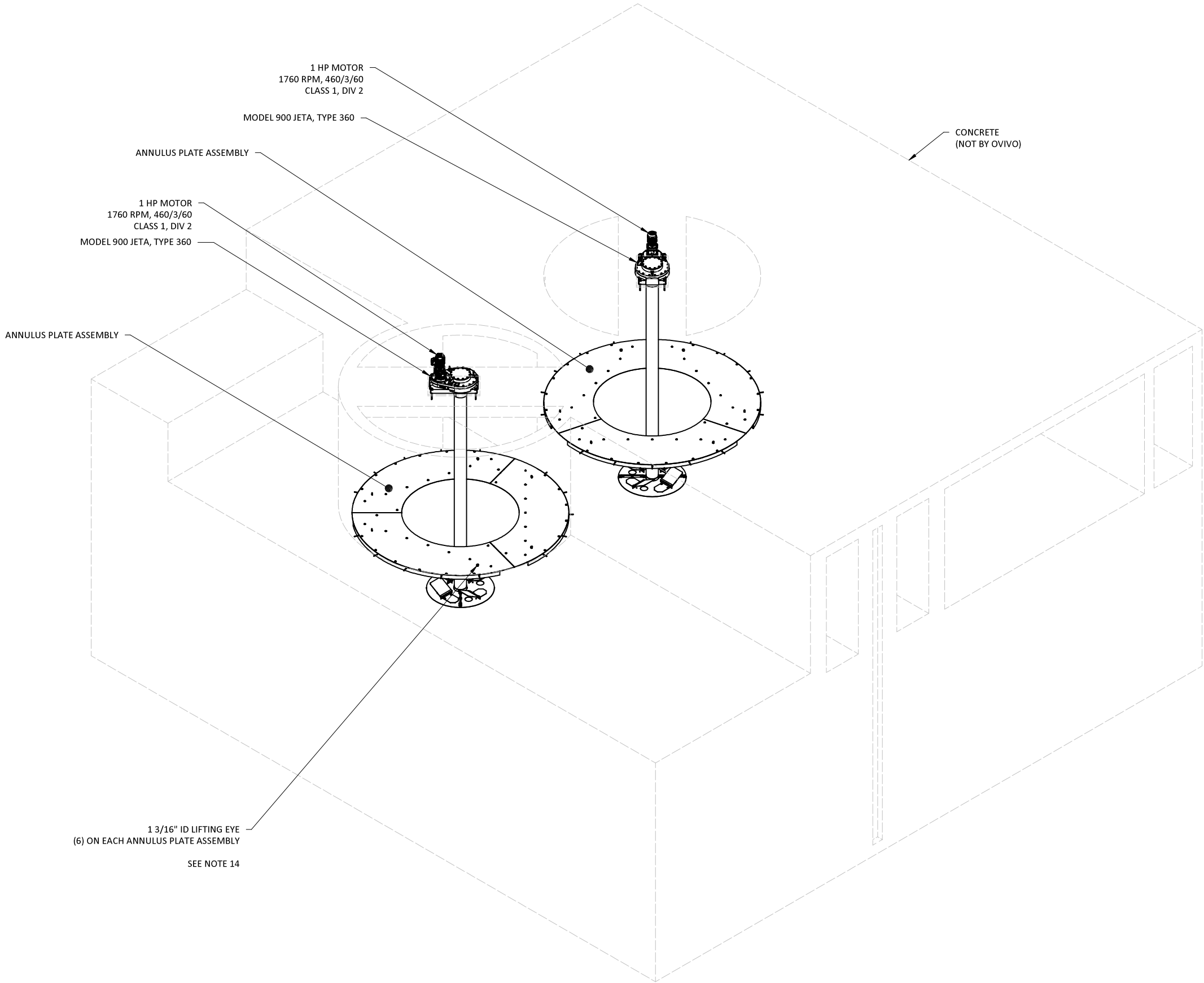
ITEM DESCRIPTION UNDER THIS COVER:

DRAWINGS.....(1) Drawing(s)

CSW0001868-02 ~ 8.22.2025

JETA MODEL DATA TABLE	
NUMBER OF GRIT CHAMBERS	2
FLUIDIZATION	WATER
WATER REQUIREMENTS	35-45 GPM @ 40-60 PSI
DESIGN FLOW	9 MGD AVG. 18 MGD PEAK

CUSTOMER TO VERIFY ALL
DIMENSIONS AND ELEVATIONS



NOTES:

1. ASTERISK (*) DENOTES VARIANCE FROM CONTRACT DOCUMENTS AND SHOULD BE PARTICULARLY NOTED.
2. THIS DRAWING REPRESENTS THE EQUIPMENT OVIVO OFFERED IN OUR PROPOSAL. ALTERATIONS OR DELAYS OF RETURN WILL AFFECT PRICE AND SCHEDULE.
3. THE FOLLOWING DEFINES THE RESPONSIBILITIES OF OVIVO WITH REGARD TO THE INFORMATION AND DIMENSIONS SHOWN ON THE DRAWING.
 - (A) DIMENSIONS, LOADS, AND OTHER INFORMATION ARE PROVIDED TO ACCOMMODATE THE EQUIPMENT TO THE STRUCTURE AS SHOWN.
 - (B) THE CUSTOMER IS TO PROVIDE REINFORCING STEEL AND DESIGN FOR CONCRETE STRUCTURES AND IS TO DETERMINE SIZES TO SUIT LOCAL CONDITIONS.
 - (C) THIS DRAWING IS NOT TO BE USED FOR CONSTRUCTION OR INSTALLATION PURPOSES.
 - (D) CHARGES FOR MODIFICATIONS, ADDITIONS OR CORRECTIONS TO THE EQUIPMENT WILL NOT BE ACCEPTED BY OVIVO UNLESS PRIOR APPROVAL IS OBTAINED IN WRITING FROM AN AUTHORIZED OVIVO REPRESENTATIVE.
4. OVIVO WILL SUPPLY
 - TWO (2) JETA GRIT COLLECTOR DRIVE W/ IMPELLER, MODEL 900/360
 - TWO (2) ANNULUS PLATE
 - TWO (2) AIRBELL
5. ANY DEMOLITION, EXCAVATION, BACKFILL, GROUT, FIELD WELDING, PIPING, WIRING, CONDUIT, INSULATION, SEALANTS, LUBRICANTS, OR FINISHING PAINTING IS NOT BY OVIVO UNLESS SPECIFICALLY NOTED.
6. ALL STRUCTURAL STEEL WILL BE 316/316L STAINLESS STEEL UNLESS OTHERWISE NOTED
7. ALL ANCHORS AND FASTENERS WILL BE 316 STAINLESS STEEL UNLESS OTHERWISE NOTED
8. ALL FABRICATED STAINLESS STEEL WILL BE CLEANED OR SAND BLASTED TO REMOVE ANY CARBON TOOLING RESIDUE AND WELD HEAT ZONE DISCOLORATION. A 24 HR WET TEST WILL BE PERFORMED TO ASTM A380.
ALL PURCHASED PAINTED ITEMS ARE TO MANUFACTURERS SPEC. I.E. MOTORS & GEARBOXES.
9. ALL WELDING WILL BE PERFORMED IN ACCORDANCE WITH THE LATEST EDITION A.W.S. WELDING PROCEDURES PER -
 - A.W.S. D1.1 - STEEL STRUCTURAL WELDING CODE.
 - A.W.S. D1.6 - STAINLESS STEEL STRUCTURAL WELDING CODE.
10. ALL FLANGES TO BE ASA B16.5 DIMENSIONS.
11. DIMENSIONS AND ELEVATIONS SHOWN ON OVIVO GENERAL ARRANGEMENT DRAWINGS HAVE BEEN TAKEN FROM THE CONSULTING ENGINEERS DRAWINGS. THESE DIMENSIONS AND ELEVATIONS ARE TO BE FIELD VERIFIED BY THE CONTRACTOR. ANY DISCREPANCIES FOUND ARE TO BE REPORTED TO OVIVO IN A TIMELY MANNER SO THAT THEY MAY BE INCORPORATED INTO THE FABRICATION OF THE OVIVO SUPPLIED EQUIPMENT. ANY CHANGES SUBSEQUENT TO THE RECEIPT OF APPROVED SUBMITTAL DRAWINGS AND SPECIFICATIONS WILL CAUSE OVIVO TO INCUR SUBSTANTIAL COSTS FOR ENGINEERING AND MATERIAL CHANGES. A CHANGE ORDER COMPENSATING OVIVO FOR THESE INCREASED COSTS WILL BE REQUIRED. OVIVO STRONGLY RECOMMENDS CAREFULLY VERIFYING ALL DIMENSIONS SHOWN ON DRAWINGS DURING THE SUBMITTAL PROCESS.
12. SPECIAL TOOLS: NONE.
13. SPARE PARTS:
 - ONE (1) DRIVE MOTOR
 - TWO (2) CHANGES OF GREASE AND OIL FOR UNITS.
14. LIFTING EYES SHALL BE REMOVED AFTER INSTALLATION TO PREVENT BINDING WITH DEBRIS.

THIS DRAWING IS CERTIFIED FOR:	
CUSTOMER:	NAN INC
CUSTOMER ORDER NO.:	24077-00036
OVIVO ORDER NUMBER:	CSW0001868-02
PROJECT:	HILO WWTP
PROJECT LOCATION:	HONOLULU, HI
CONSULTING ENGINEER:	-
BY: ALEX WANG	DATE: 04/11/2025

DRAWN BY
CHECKED BY
DATE
PROJECT
REF

NM
AW
04/11/2025
CSW0001868-02
CSW1699-101

GENERAL ARRANGEMENT
GRIT REMOVAL SYSTEM
HILO WWTP, HI

CSW1868-201

D SIZE SHEET

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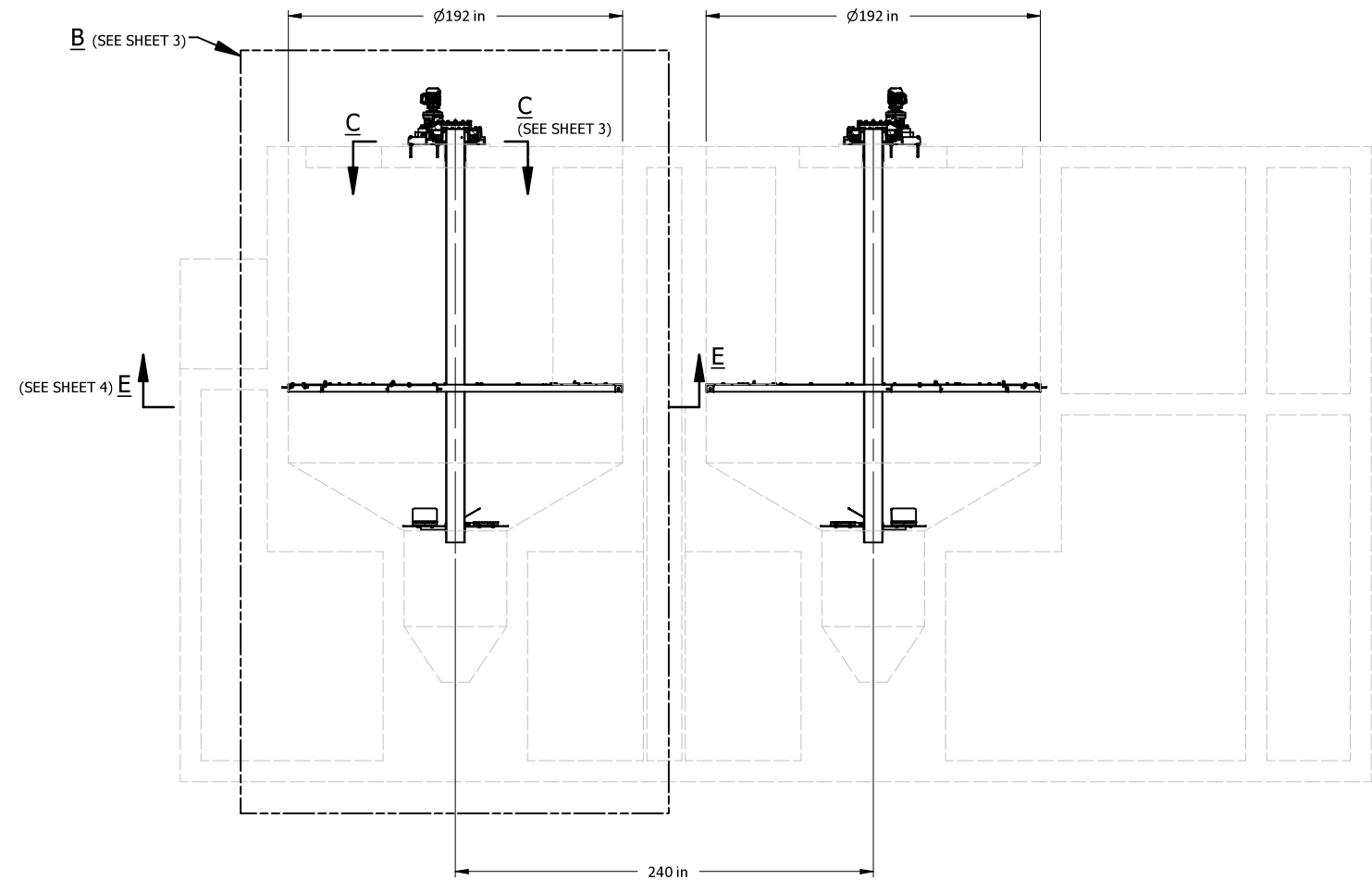
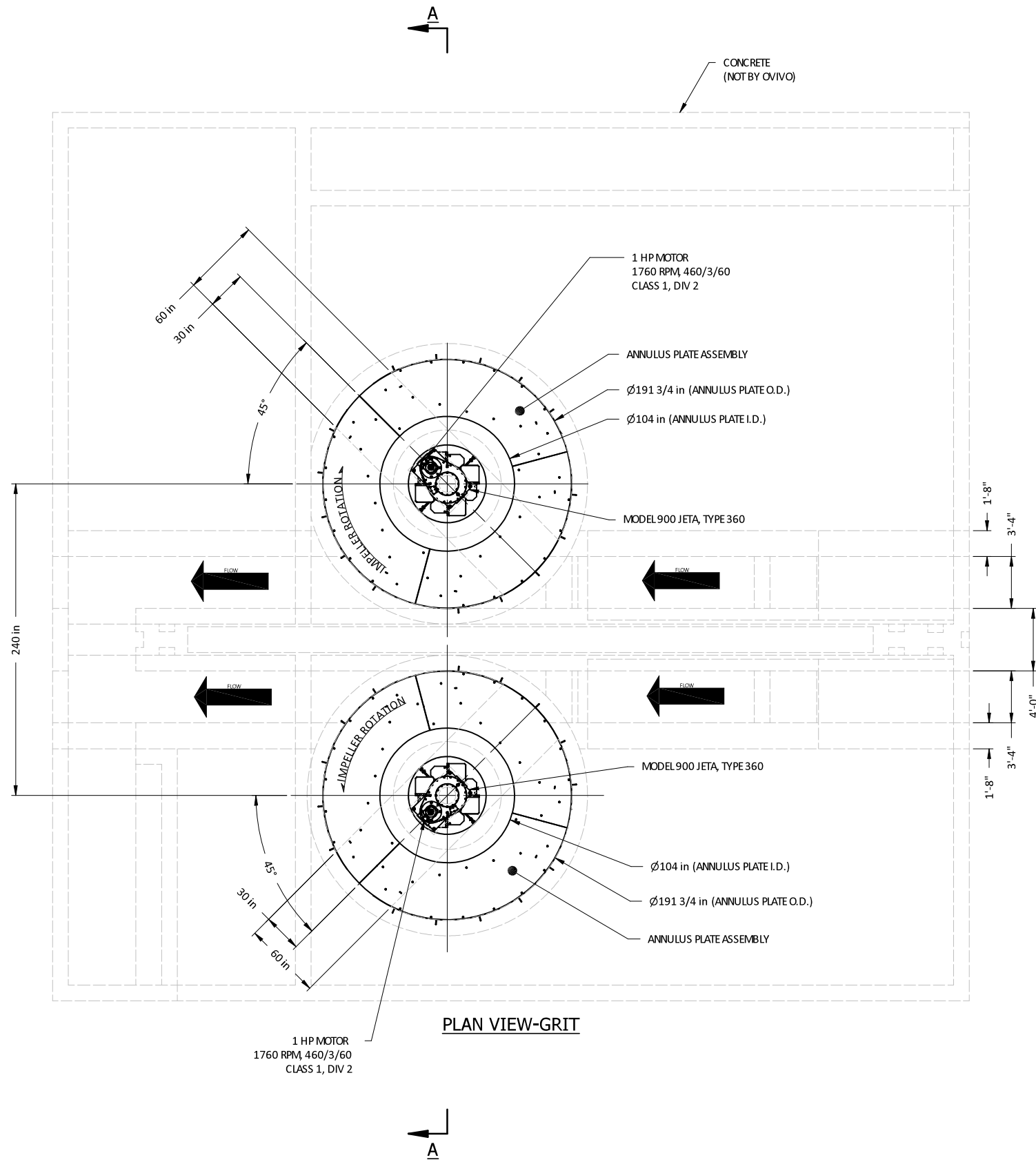
SHEET 1 OF 4

REVISION A

ISOMETRIC VIEW

INITIAL RELEASE	REVISION SUMMARY	ECO	-	-	DATE	A
		BY	CHECKED		REV	

CUSTOMER TO VERIFY ALL
DIMENSIONS AND ELEVATIONS



D SIZE
SHEET
ISO 15926-2:2018 (2018) (2018)
(2018) (2018) (2018)

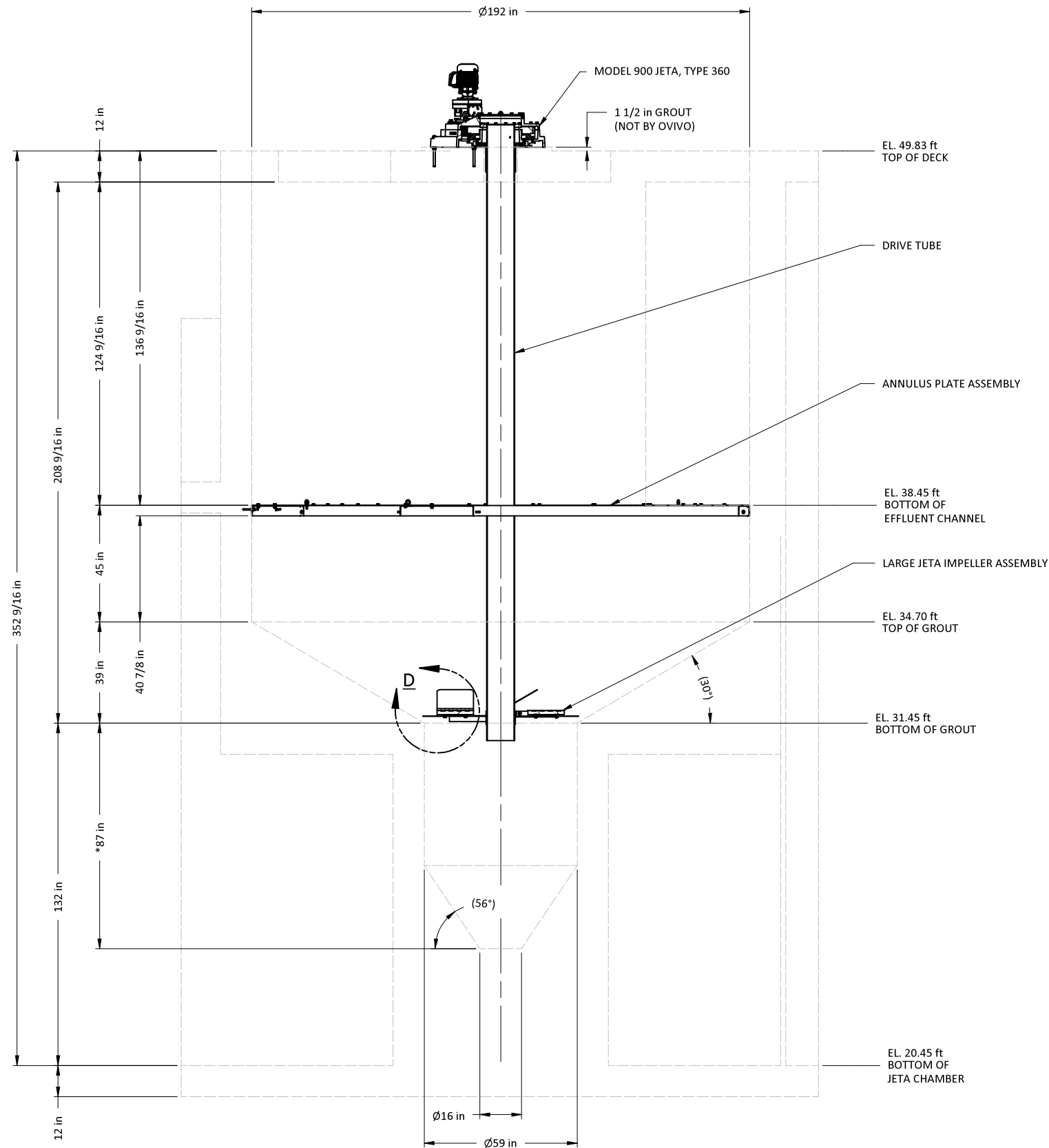
CSW1868-201

SHEET
2 OF 4
A
REVISION

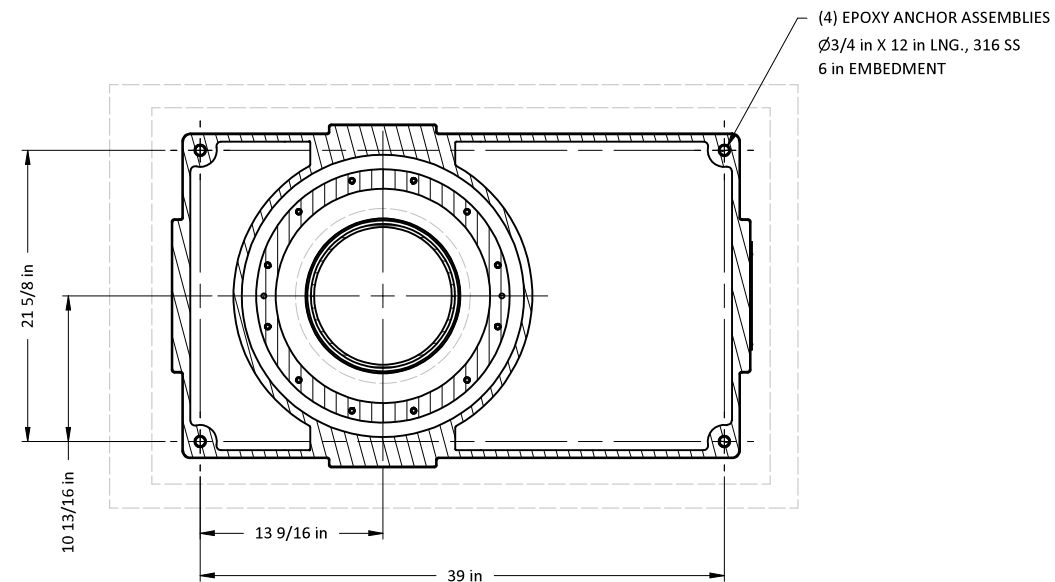
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CUSTOMER TO VERIFY ALL
DIMENSIONS AND ELEVATIONS

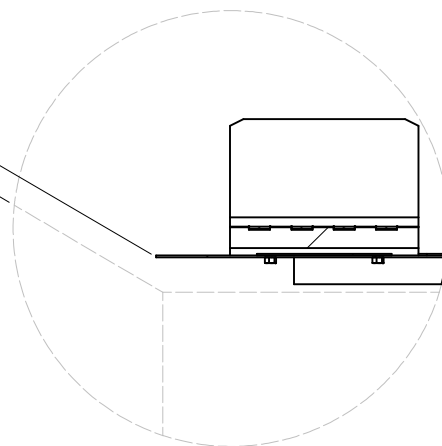


DETAIL B
(FROM SHEET 2)



SECTION C-C
(JETA DRIVE HEAD ANCHOR LAYOUT)
(FROM SHEET 2)

2 in GAP (CRITICAL DIMENSION)
ADJUST HEIGHT OF IMPELLER AS NECESSARY
TO ACHIEVE 2 in GAP SHOWN



DETAIL D

CUSTOMER TO VERIFY ALL
DIMENSIONS AND ELEVATIONS

(15) Ø1/2 in X 6 in LNG.
EPOXY ANCHOR ASSEMBLY, 316 SS
4" MIN. EMBED

Ø191 3/4 in
ANNULUS PLATE ASSEMBLY

SECTION E-E
(ANNULUS PLATE DETAIL VIEW)
(FROM SHEET 2)



D SIZE
SHEET

CSW1868-201

SHEET
4 OF 4

A
REVISION

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